

Coarse Revenue Guarantee in First Price Auction and Beyond*

Linchen Zhang[†]

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Abstract

I analyze the coarse revenue guarantee (CRG) of the single-unit first-price auction with common values: CRG is the minimum expected revenue over Bayes Coarse Correlated Equilibria (BCCE), and BCCE is the equilibrium notion that no-regret learning leads to. The worst case is identical play, in which all bidders pool on a common bid, and this reduction yields a closed-form characterization. The guarantee is always positive, increasing in the number of bidders, and strictly below the interim guarantee of [Bergemann, Brooks and Morris \(2017\)](#), converging to it at rate $\log N/N$. The analysis is then extended to reserve prices, symmetric priors, other standard auctions beyond the first-price auction, and unknown value distributions.

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[†]Department of Economics, University of California San Diego. Email: liz032@ucsd.edu.

1 Introduction

The robustness of auction mechanisms to the information structure is a central question in mechanism design. A seller who knows the distribution of the good’s value but not the correlated information structure between all buyers may want a lower bound on revenue that holds across every environment consistent with that prior. [Bergemann, Brooks and Morris \(2017\)](#) is a seminal work that provides such a bound for the first-price auction. They characterize the **revenue guarantee** (RG), defined as the minimum expected revenue across all information structures and all Bayes Correlated Equilibria (BCE), and show that it is positive, even as the number of buyers grows large.

However, the BCE framework has the Bayesian assumption. In a BCE, a bidder who is told to submit a_i after observing signal s_i must be unwilling to change that bid *at that signal*. Obedience holds signal by signal, which requires each bidder to process their private information and best-respond conditional on it. This is the natural requirement when bidders are sophisticated Bayesian agents.

But it may not be the appropriate requirement when bidders are algorithms. In modern electronic marketplaces, and increasingly in advertising exchanges and high-frequency trading venues, bids are submitted by automated agents. These agents adjust their bids over time by simple learning rules rather than by solving a Bayesian inference problem. The standard performance criterion for such rules is *no regret*: the algorithm guarantees that, in hindsight, no single fixed action played throughout the history would have earned more on average. That is an unconditional guarantee. It says nothing about whether the algorithm’s behavior would have been optimal in any particular state, and a no-regret algorithm can have large regret conditional on individual states without ever violating its guarantee.¹

The solution concept that addresses this weaker notion of rationality is Bayes Coarse Correlated Equilibrium (BCCE), in which each player must be unwilling to switch to a single fixed action played conditional on every signal. [Hartline, Syrgkanis and Tardos \(2015\)](#) show that the empirical distributions generated by no-regret dynamics in a Bayesian game accumulate on the set of BCCE, not on the smaller set of BCE. A seller facing algorithmic bidders who want a lower bound on revenue

¹See [Lomys and Magnolfi \(2025\)](#) for an econometric framework built on exactly this weaker restriction, and [Nekipelov, Syrgkanis and Tardos \(2015\)](#) for inference from no-regret bidders.

that holds for every outcome those algorithms can converge to must therefore minimize over BCCE. This paper carries out a similar exercise of [Bergemann, Brooks and Morris \(2017\)](#) for BCCE.

In this paper, I define the **coarse revenue guarantee** (CRG) as the minimum expected revenue across all BCCEs, and I ask four questions. How large is it in the first-price auction and when is it attained? How much does the seller lose by facing coarsely rational bidders rather than interim rational ones?² What can the seller do to raise it? And how do the results change across settings, namely symmetric priors, other auction formats, and an unknown value distribution?

Characterization through identical play The first contribution is a tractable characterization, and it is built on a structural result. In the first-price auction, from any BCCE one can construct another BCCE with weakly lower revenue, in which every buyer submits the same bid with probability one. The worst case is then an *identical play*: bidders do not use their private information and all submit the same bid. This construction works because of two effects that act in the same direction. First, revenue in a first-price auction is the highest bid, so replacing a dispersed bid profile with a common bid that is no higher than the highest bid in the profile lowers revenue. This effect holds under any notion of rationality. Second, the replacement can be arranged so that the distribution of the *highest competing bid* faced by any individual buyer is unchanged. This is why the construction preserves the equilibrium constraints. A coarse deviation is a single fixed bid, and the profitability of a fixed bid depends on one's rivals only through the distribution of the best bid among them. Holding that distribution fixed while lowering the price means that each buyer's payoff from deviating is unchanged while their payoff from obeying goes up. The constraints are thus relaxed, not violated.

The structure of this worst case is where the coarse analysis departs from the BCE benchmark. In [Bergemann, Brooks and Morris \(2017\)](#) the worst case is supported by an information structure in which bidders learn something about their rank, and play symmetrically *conditional* on that information. Under coarse rationality, the

²The word *interim* refers to timing and is used throughout as the counterpart of *coarse*. In a Bayes correlated equilibrium a bidder decides whether to deviate after seeing their information, so the constraint is checked at the interim stage, conditional on the signal and the recommended bid. In a coarse equilibrium the deviation is one fixed bid chosen before any information arrives, so the constraint is ex ante, an average over all states. Section 3 gives the formal definitions.

binding force is instead unconditional identical play. Bidders who cannot condition a deviation on their signal can be induced to submit a common bid that does not depend on the signal.

In terms of tractability, the reduction turns an N -dimensional problem into a one-dimensional one. Instead of searching over joint distributions of values, signals, and N -dimensional bid profiles, the seller's worst case is now determined by the joint distribution of the value v and a single common bid c . Theorem 1 solves this problem in closed form for an arbitrary value distribution. In the worst case, buyers bid zero when the value is below a threshold, so the good is sold at a price of zero there, and bid a common increasing amount above the threshold. The surplus from the zero-bid region is what keeps buyers from deviating to a slightly higher bid. The guarantee is strictly positive for every non-degenerate prior and strictly increasing in the number of buyers.

The cost of coarseness Because every BCE is a BCCE, the coarse guarantee is weakly below the interim guarantee of Bergemann, Brooks and Morris (2017), and Proposition 4 shows that the inequality is strict at every finite N , since no BCE ever attains the coarse minimum. The reason is that the worst case has a rigid structure, and that structure cannot satisfy interim obedience. Quantitatively, the gap is large when the number of bidders is small.

Nevertheless, Theorem 2 shows that this gap closes as N grows. The coarse constraint is satisfied by lowering revenue until each buyer's equilibrium payoff exceeds every fixed deviation's payoff, and with N buyers sharing the surplus, that payoff is only $1/N$ of the total. As N grows, the revenue reduction needed to satisfy the coarse constraint goes to zero, and the two guarantees coincide in the limit. Proposition 3 shows, however, that the convergence is slow. The interim guarantee reaches the common limit at rate $1/N$, the coarse guarantee only at rate $\log N/N$, so the cost of coarseness $RG_N - CRG_N$ itself vanishes at rate $\log N/N$.

What the seller could do The next question is what the seller can do to increase the coarse guarantee. One natural instrument is a reserve price, under which the good is withheld if the highest bid is below the reserve. Section 6 shows that the identical play reduction logic largely holds with a reserve price, with a modified pooling construction. The same tools still apply, and the guarantee is then characterized at every

reserve level. Two findings then follow. First, a small reserve raises the guarantee only slightly. It might be tempting to expect that any positive reserve can completely eliminate the states in which bidders receive the good at a price of zero and restore it perfectly. But then the buyers have a cheaper response: they can just pool on a common bid at r , the reserve price. And it is indeed the worst case if r is small. Second, a larger reserve may raise the guarantee substantially, but backfire after a point. As the reserve grows, pooling at the reserve becomes costly for the buyers, so the worst case gradually shifts from pooling at the reserve to always not buying for a very large reserve price.

Symmetric Priors The analysis in this paper is set in a pure common value model. Since the interim characterization of [Bergemann, Brooks and Morris \(2017\)](#) covers symmetric priors with arbitrary correlation, this setting is more restrictive, and readers may want to ask what would be different if relaxing to symmetric priors. Section 7 examines what changes when values may differ across buyers. The first change is that identical play is no longer without loss. When values differ, the adversary aiming to reduce the seller’s revenue has an additional instrument. It can keep the price common and still allocate the good to the buyer with the highest value, by an arbitrarily small bid premium, and this change still satisfies the obedience constraints. Nevertheless, Theorem 4 shows that the guarantee for any exchangeable prior is bounded below by the fixed point of Theorem 1 with the mean value replaced by the expected highest value, and that the bound is exact whenever values agree above the resulting threshold. Whenever values differ within the pooling region, the guarantee is strictly below the common value guarantee. The worst symmetric priors therefore agree at high values and differ at low ones.

Ranking and equivalence across auctions Section 8 compares the first-price auction with the other standard auctions, following [Bergemann, Brooks and Morris \(2019\)](#). The comparison is based on one observation. Under identical play, every standard auction allocates by the same tie-breaking rule and asks for the same price, so the outcome attaining the first-price guarantee is available in every format, and formats differ only in what a single deviator pays after outbidding the others. When that payment is the deviator’s own bid, the identical play equilibria of the first-price auction are equilibria of the format. Its guarantee restricted to identical play is the

first-price guarantee, and its full guarantee is at most that. When the deviator pays the others' bid instead, as in the second-price and English auctions, outbidding them does not raise the price paid, pooling at a low price cannot be sustained, and an asymmetric equilibrium yields a guarantee of zero. The first-price auction has the highest guarantee in both cases, so a seller designing for no-regret bidders is led to the same format as a seller designing for Bayesian bidders.

Unknown value distribution Finally, Section 9 considers a seller who knows only the mean of the value distribution and wants a lower bound on revenue over BCCE. I first show that the coarse revenue guarantee is monotone in the convex order: A prior that is mean-preserving-spread of another is worse for the seller. Because the binary distribution on $\{0, \bar{v}\}$ is the most spread-out distribution on $[0, \bar{v}]$ with a given mean, it is the worst-case prior, and its guarantee provides another lower bound. I then solve for this bound in closed form.

The remainder of the paper is organized as follows. Section 2 discusses related literature. Section 3 introduces the model and the two equilibrium notions. Section 4 establishes the identical play reduction. Section 5 characterizes the coarse revenue guarantee and establishes asymptotic convergence and its rate. Section 6 develops the reserve-price analysis. Section 7 extends the characterization to symmetric priors. Section 8 compares the guarantee across standard auctions. Section 9 treats an unknown value distribution. Section 10 concludes. Omitted proofs are in the Appendix.

2 Related Literature

This paper contributes to three strands of literature: first-price auctions, robust revenue guarantees, and the intersection of Bayes Coarse Correlated Equilibrium with no-regret learning dynamics.

First, this work builds on the literature on the first-price auction. Early studies, such as Vickrey (1961), Milgrom and Weber (1982), Battigalli and Siniscalchi (2003), Dekel and Wolinsky (2003), Fang and Morris (2006), and Azacis and Vida (2015), relied on specific assumptions about the information structure. More recently, Bergemann, Brooks and Morris (2017) allowed for arbitrary information structures but restricted attention to Bayes Correlated Equilibrium. This paper characterizes

the coarse revenue guarantee under the weaker notion of BCCE, and by developing coarse counterparts of the ranking and equivalence results of [Bergemann, Brooks and Morris \(2019\)](#). The closest earlier work is [Feldman, Lucier and Nisan \(2016\)](#), who study correlated and coarse equilibria of single-item auctions. Their focus is on welfare in the full-information model, whereas I minimize revenue over all information structures and characterize the minimizing equilibrium exactly.

Second, this paper contributes to the literature on revenue guarantees, including [Bergemann, Brooks and Morris \(2017\)](#), [Bergemann, Brooks and Morris \(2019\)](#), [Du \(2018\)](#), and [Brooks and Du \(2021\)](#), by investigating these guarantees outside the scope of BCE. While there is existing work studying guarantees in terms of welfare, specifically the price of anarchy ([Syrkanis and Tardos, 2013](#); [Jin and Lu, 2023](#)), this paper focuses strictly on revenue rather than welfare outcomes. The reserve-price result of Section 6 also connects this paper to the literature comparing competition with mechanism design initiated by [Bulow and Klemperer \(1996\)](#), whose theorem states that an auction with one additional bidder dominates the optimal mechanism with a reserve price. Against coarsely rational bidders a small reserve raises the guarantee only slightly, because bidders pool at the reserve and pay it. A large gain requires a reserve large enough that the worst case involves withholding rather than pooling at the reserve.

Finally, this work contributes to the literature on BCCE and learning in games. BCCE was first proposed by [Moulin and Vial \(1978\)](#). It received little attention in economics because its set of deviations is small. The use of no-regret algorithms as bidders has renewed interest. As shown by [Hartline, Syrgkanis and Tardos \(2015\)](#) and supported by [Hart and Mas-Colell \(2000\)](#) and [Blum et al. \(2008\)](#), no-regret dynamics converge to the set of BCCE. A growing literature studies auctions in which some or all participants are learning algorithms: [Nekipelov, Syrgkanis and Tardos \(2015\)](#) on inference from no-regret bidders, [Braverman et al. \(2018\)](#) and [Deng, Schneider and Sivan \(2019\)](#) on a seller’s ability to exploit them, [Kolumbus and Nisan \(2022\)](#) on auctions between learning agents, and [Lomys and Magnolfi \(2025\)](#) on structural estimation under no-regret restrictions. Relative to that literature, which typically fixes a learning protocol or an information structure, I take the set-valued approach of robust mechanism design and bound revenue over the entire BCCE set.

3 Model

3.1 Setup

Consider a single-unit auction with a set of risk-neutral buyers $\mathcal{N} = \{1, \dots, N\}$, where $N \geq 2$. The seller owns one indivisible good and values it at zero. The buyers have a pure common value for the good, denoted $v \in V = [0, \bar{v}] \subseteq \mathbb{R}_+$, distributed according to a probability measure $\mu \in \Delta(V)$ with mean $m := \mathbb{E}_\mu[v]$ and cdf F . Section 7 treats general symmetric priors, for which the common value coupling is the worst case within identical play but not overall.

Assumption 1. *μ admits a density f that is continuous and positive on $[0, \bar{v}]$, and each buyer's action set is $A_i = [0, \bar{v}]$, so that $A = \prod_i A_i = [0, \bar{v}]^N$.*

The density assumption is kept throughout to simplify the exposition, not because the results need it, except for Proposition 3, which uses the continuity and positivity of f at 0. Under it F is continuous and strictly increasing on $[0, \bar{v}]$, and F^{-1} denotes its inverse. Section 9 extends the characterization to arbitrary priors, including atomic ones, which is needed there because the worst-case prior among distributions with a given mean is binary.

Since a bid above $\bar{v}$ guarantees a loss whenever it wins in the first-price auction, restricting bids to $[0, \bar{v}]$ is without loss. Later on, in Section 8, where the set of allowed deviations is exactly what distinguishes one format from another, it may not be without loss. Hence, I state the restriction explicitly.

In the first-price auction (FPA), the highest bidder wins. If there are ties, the good is allocated with equal probability among the tying bidders. Formally, the allocation rule $q_i(a)$ for buyer i is

$$q_i(a) = \begin{cases} 1, & \text{if } a_i > \max_{j \neq i} a_j, \\ \frac{1}{|\{k \in \mathcal{N} : a_k = \max_j a_j\}|}, & \text{if } a_i = \max_j a_j, \\ 0, & \text{otherwise,} \end{cases}$$

and the winner pays their own bid. Buyer i 's ex-post utility given the common value v and the action profile a is

$$u_i(v, a) = q_i(a)(v - a_i).$$

Moreover, since the good is always allocated, the following equation must be true:

$$\sum_i u_i(v, a) = v - \max_i a_i \quad \text{for every } (v, a). \quad (1)$$

Anything that lowers revenue would raise the surplus of the buyers by exactly the same amount.

3.2 Information and Two Notions of Obedience

The buyers' correlated information about the common value is described by an *information structure* $I = (S, \theta)$, where S_i is a finite set of signals for buyer i , $S = \prod_i S_i$, and $\theta \in \Delta(V \times S)$ is the joint distribution of values and signals. Let $\mathcal{I}(\mu)$ be the set of information structures whose value marginal is the prior, $\text{marg}_V \theta = \mu$. The seller does not know I , and the guarantee is computed over all $I \in \mathcal{I}(\mu)$.

Given I , a *decision rule* is a map $b : S \rightarrow \Delta(A)$. Equivalently, a mediator observes the signal profile s and privately recommends a bid a_i to each buyer i . A decision rule induces the joint distribution

$$\sigma_b(dv, ds, da) = b(da \mid s) \theta(dv, ds) \in \Delta(V \times S \times A),$$

and buyer i 's expected utility is $U_i(I, b) := \int u_i(v, a) \sigma_b(dv, ds, da)$.

The two solution concepts I compare differ in exactly one respect: *how much a buyer is allowed to know when choosing a deviation*. I state both in turn.

Definition 1 (Bayes Correlated Equilibrium). *A decision rule b is a Bayes correlated equilibrium of I if for every buyer i and every measurable map $\xi_i : S_i \times A_i \rightarrow A_i$,*

$$\int u_i(v, a) \sigma_b(dv, ds, da) \geq \int u_i(v, \xi_i(s_i, a_i), a_{-i}) \sigma_b(dv, ds, da).$$

Write $\mathcal{BCE}(I)$ for the set of BCE of I .

Definition 1 is the standard obedience formulation of [Bergemann and Morris \(2016\)](#). The recommended strategy $a_i = b(s_i)$ is optimal given the signal s_i . Obedience must hold conditional on signals. The map ξ_i may use a different deviation a'_i for another signal s'_i .

Definition 2 (Bayes Coarse Correlated Equilibrium). *A decision rule b is a Bayes coarse correlated equilibrium of I if for every buyer i and every constant deviation $a'_i \in A_i$,*

$$\int u_i(v, a) \sigma_b(dv, ds, da) \geq \int u_i(v, a'_i, a_{-i}) \sigma_b(dv, ds, da).$$

Write $\text{BCC}\mathcal{E}(I)$ for the set of BCCE of I .

By contrast, in a BCCE the buyer only checks fixed-action deviations. The no-regret learning algorithm converges to this set rather than to the smaller BCE set (Hartline, Syrgkanis and Tardos, 2015; Blum et al., 2008). In the language used throughout the paper, Definition 1 imposes *interim* obedience, evaluated after the buyer sees their information and so that conditional on signals, while Definition 2 imposes *ex ante*, or coarse, obedience, evaluated before, so that it is unconditional on signals.

Because a constant map is one particular instance of a deviation map ξ_i , we have $\text{BCE}(I) \subseteq \text{BCC}\mathcal{E}(I)$ for every I : BCCE is the more permissive concept. The following example illustrates the difference.

Example 1 (A BCCE that is not a BCE). Let $N = 2$ and let v take the values 0 and 1 with equal probability. Suppose both buyers observe v and the mediator recommends the common bid $c = v/3$, so that both bid 0 when $v = 0$ and both bid $1/3$ when $v = 1$. Each buyer's expected payoff is $\frac{1}{2} \cdot \frac{1}{2} \cdot (1 - \frac{1}{3}) = 1/6$. A buyer deviating to the fixed bid $\delta \in (0, 1/3)$ wins only when $v = 0$, earning $\frac{1}{2}(0 - \delta) < 0$. A buyer moving to $\delta \in (1/3, 1)$ always wins and earns $\frac{1}{2}(0 - \delta) + \frac{1}{2}(1 - \delta) = \frac{1}{2} - \delta < \frac{1}{6}$. A buyer deviating to exactly $1/3$ wins when $v = 0$ and ties when $v = 1$, earning $\frac{1}{2}(0 - \frac{1}{3}) + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{2}{3} = 0$. Hence, no fixed deviation generates more payoff than $1/6$, so the profile is indeed a BCCE. However, it is not a BCE, because conditional on the recommendation $c = 1/3$, the buyer knows $v = 1$ and can deviate to $1/3 + \epsilon$ to win, and that results in a higher payoff than the BCCE given ϵ small.

Example 1 illustrates the mechanism behind the paper's results. Under coarse rationality, revenue can be lowered by equilibria in which bidders would gain from deviating in the states where the good is valuable, but are deterred by the losses they would incur in the states where it is not. The low bid is sustainable because the deviation payoff is averaged across states, and interim obedience does not average across states.

Remark 1 (Between coarse and interim obedience). The two definitions are the extreme cases of a family, ordered by how finely a deviation may condition on information. Fix for each buyer a partition of their signal set and require obedience against deviations that are constant on each cell. Coarsening every partition to a single cell is Definition 2, while the finest partition using every measurable signal is Definition 1. The equilibrium sets are nested and become smaller as partitions are finer, so worst-case revenue is monotone along the family, and the two guarantees studied in this paper are the two extremes. Contextual algorithms take value-relevant features as inputs and result in a solution concept that is in the middle of BCE and BCCE. As a result, the per-context guarantee lies strictly between the two, and the coarse revenue guarantee can provide a valid lower bound for the per-context guarantee as well.

With the two equilibrium notions in place, let $R(I, b) := \int \max_i a_i \sigma_b(dv, ds, da)$ be expected revenue. The *revenue guarantee* and the *coarse revenue guarantee* under the prior μ are

$$RG(\mu) := \inf_{I \in \mathcal{I}(\mu)} \inf_{b \in \mathcal{BCE}(I)} R(I, b), \quad CRG(\mu) := \inf_{I \in \mathcal{I}(\mu)} \inf_{b \in \mathcal{BCCCE}(I)} R(I, b),$$

and since minimizing over a larger set yields a smaller value, $CRG(\mu) \leq RG(\mu)$ for every prior.

3.3 A Reduced Notation for the Coarse Problem

Definition 2 has a property that Definition 1 lacks. Since a coarse deviation is unconditional on the signal, the signals would never appear in the constraint except for the joint distribution they induce over values and actions. The information structure can therefore be, without loss, replaced by a joint distribution $\sigma \in \Delta(V \times A)$ whose value marginal is μ .

Lemma 1 (Reduced formulation). *A distribution $\sigma \in \Delta(V \times A)$ with $\text{marg}_V \sigma = \mu$ is the value-action marginal of some σ_b with $I \in \mathcal{I}(\mu)$ and $b \in \mathcal{BCCCE}(I)$ if and only if, for every buyer i and every $a'_i \in A_i$,*

$$\int_{V \times A} u_i(v, a) d\sigma(v, a) \geq \int_{V \times A} u_i(v, a'_i, a_{-i}) d\sigma(v, a). \quad (2)$$

Proof. Necessity holds because both sides of Definition 2 depend on σ_b only through its $V \times A$ marginal. For sufficiency, let the signal be the null signal so that every buyer has no additional information other than the prior, and let the mediator recommend the bid profile from $\text{marg}_A \sigma(\cdot \mid v)$. The induced joint law is σ . (2) is then the BCCE equilibrium constraint. $\square$

Write $\mathcal{BCC}\mathcal{E}(\mu)$ for the set of σ satisfying (2), and $R(\sigma) = \int \max_i a_i d\sigma$. Then

$$CRG(\mu) = \inf_{\sigma \in \mathcal{BCC}\mathcal{E}(\mu)} R(\sigma),$$

and any σ attaining this infimum is called a *min-R BCCE*.

Note that I do not assume that such a σ exists. Because the payoff functions u_i are discontinuous at ties, the map $\sigma \mapsto \int u_i d\sigma$ is not continuous in the weak-* topology, and $\mathcal{BCC}\mathcal{E}(\mu)$ need not be weak-* closed, so the existence of a minimizer does not follow from compactness. Every argument below is therefore constructive. Each step maps a given BCCE to another BCCE with weakly lower revenue, and existence of the minimizer is established at the end by exhibiting it.

Ties also require a separate argument, since the payoff functions are discontinuous there. The following lemma lets every subsequent argument ignore them. It says that for an identical play distribution it is enough to check the coarse constraint at bids which the buyers do not submit with positive probability, and these are dense.

Lemma 2 (Tie-free verification). *Let c be a random common bid submitted by all N buyers, and let K be each buyer's equilibrium payoff. Suppose that*

$$K \geq \mathbb{E}[(v - \delta) \mathbb{1}\{c < \delta\}]$$

for every δ in a dense subset $D \subseteq [0, \bar{v}]$. Then the coarse obedience constraint (2) holds at every $\delta \in [0, \bar{v}]$, ties included.

The proof is in Appendix A.

4 The Identical Play Reduction

Minimizing revenue directly over $\mathcal{BCC}\mathcal{E}(\mu)$, a set of distributions over N -dimensional bid profiles, is not tractable. This section shows that the minimization can be restricted to a one-dimensional family. Call σ an *identical play* distribution if it puts

probability one on profiles in which all buyers submit the same bid, $a_1 = \dots = a_N$. Call a distribution over (v, a) *exchangeable* if it is invariant under every permutation of the buyers' bids, $(v, a) \mapsto (v, a_\rho)$ with $(a_\rho)_i = a_{\rho(i)}$.

Proposition 1 (Identical Play Reduction). *For every $\sigma \in \mathcal{BCCE}(\mu)$ there is an exchangeable identical play $\gamma \in \mathcal{BCCE}(\mu)$ with $R(\gamma) \leq R(\sigma)$, and the inequality is strict unless σ already places probability one on profiles whose two highest bids coincide. Consequently*

$$CRG(\mu) = \inf \{R(\gamma) : \gamma \in \mathcal{BCCE}(\mu), \gamma \text{ identical play}\}.$$

The formal argument is in Appendix A. The main idea is that in the first-price auction, the price competition comes from the buyer with the highest bid and the buyer with the second-highest bid. When a person needs to check the obedience constraints, she needs to optimize against the distribution of the highest bid among others. The lower revenue argument of the identical play comes from in some cases, for a bid profile, it is possible to replace the highest bid with the second-highest bid and still keep the distribution of competition the same, and therefore still satisfy the obedience constraint. As a result, it yields a BCCE with lower revenue. This revenue reduction channel is only impossible if the highest bid and the second-highest bid are the same with probability one. Moreover, since the price competition is only between the person with the highest bid and the second-highest bid, all the rest of the bids play no role, and it is without loss to let them be at the same level of the second-highest bid. Hence, the new BCCE construction can be viewed as an identical play.

With this reduction in hand, the remaining problem is to choose the joint distribution of the value v and a single common bid c .

5 Characterization

5.1 The Representative-Bidder Program

Fix an identical play $\gamma \in \mathcal{BCCE}(\mu)$, write c for the common bid and $R = \mathbb{E}[c]$ for expected revenue. Since all N bids coincide, every buyer wins with probability $1/N$

and pays c when they do, so each buyer's equilibrium payoff is

$$K := \frac{m - R}{N}, \quad (3)$$

the per-buyer share of the surplus not collected by the seller. I refer to K as the buyers' *rent*. A buyer who instead commits to the fixed bid δ wins whenever $c < \delta$ and shares when $c = \delta$, so coarse obedience requires, for every $\delta \in [0, \bar{v}]$,

$$K \geq \mathbb{E} \left[(v - \delta) \left(\mathbb{1}\{c < \delta\} + \frac{1}{N} \mathbb{1}\{c = \delta\} \right) \right]. \quad (4)$$

Setting $\delta = 0$ shows $K \geq 0$: revenue can never exceed the expected value.

Before stating the main result, I note two properties of the equilibrium constraint that shape it.

First, lowering revenue relaxes the constraints. (3) says that lowering R raises K , and K appears on the left-hand side of every constraint (4). The guarantee is then related to a K that makes the equilibrium constraint binding.

Second, a small deviation only wins for items where everyone else bids zero. Since each buyer has a payoff of K by following the equilibrium, the small deviation that gives a payoff less than K is not profitable. Therefore, for the low value regions of the item, it is possible that buyers jointly bid 0 and get the item for free, and the positive bidding region begins only above a value threshold. The following definition names the value at which the bidding region begins.

Definition 3 (Threshold and worst-case bidding rule). *Let $\hat{v}$ denote the pooling threshold: in the worst-case equilibrium, buyers bid zero whenever $v < \hat{v}$ and bid a strictly increasing amount above it. Write $H(v)$ for the common bid submitted at value $v \geq \hat{v}$.*

5.2 Main Result

Theorem 1. *Let Assumption 1 hold. The coarse revenue guarantee CRG and the threshold $\hat{v}$ are the unique solution of the system*

$$\frac{m - CRG}{N} = \int_0^{\hat{v}} v f(v) dv, \quad (5)$$

$$CRG = \int_0^{\hat{v}} H(v) f(v) dv, \quad (6)$$

where $H(v) = 0$ for $v < \hat{v}$ and, for $v \geq \hat{v}$,

$$H(v) = \frac{1}{F(v)} \int_{\hat{v}}^v z f(z) dz = \mathbb{E}[z \mid \hat{v} \leq z \leq v] \cdot \frac{F(v) - F(\hat{v})}{F(v)}. \quad (7)$$

The guarantee is strictly positive and can be attained: the identical play distribution with common bid $c = H(v)$ is a BCCE achieving it.

Appendix B contains the proof. There are three main steps. A *lower bound* shows that no identical play BCCE can do better than the system above. This comes from combining the obedience constraint with a rearrangement inequality, rather than from assuming that constraints bind. *Sufficiency* verifies that the candidate rule (7) deters every fixed deviation, including deviations above the top of the bid range. The last step is the *uniqueness* of the solution to (5)–(6). This step is not immediate, since both sides of the system move in the same direction as $\hat{v}$ rises. It follows from a concavity argument given in the appendix.

The fixed-point form. For later use, I record an equivalent form of the system, written in the quantile variable $q = F(v)$. Let

$$\Phi_\mu(q) := \int_0^q F^{-1}(t) dt = \int_0^{F^{-1}(q)} z f(z) dz, \quad q \in [0, 1], \quad (8)$$

the expected value collected on the lowest q quantiles of the prior. It is convex and strictly increasing, with $\Phi_\mu(0) = 0$ and $\Phi_\mu(1) = m$. Define the revenue functional

$$\mathcal{R}_\mu(K) := \int_0^1 \frac{(\Phi_\mu(q) - K)^+}{q} dq, \quad K \in [0, m]. \quad (9)$$

The system (5)–(6) is equivalent to the single equation

$$m - NK = \mathcal{R}_\mu(K) \quad (10)$$

in the rent: the unique solution is $K^* = \int_0^{\hat{v}} v f(v) dv$, the guarantee is $CRG = m - NK^* = \mathcal{R}_\mu(K^*)$, and the worst-case bid is $c = \psi(F(v)) = H(v)$, where

$$\psi(q) := \frac{(\Phi_\mu(q) - K^*)^+}{q}. \quad (11)$$

Appendix B proves the theorem in this form. The functional $\mathcal{R}_\mu$ is also used in the reserve-price analysis of Section 6 and in the convex-order comparison of Section 9.

Equation (5) is the participation condition. Its right-hand side is the expected value collected on the pooling region. It also equals the payoff from deviating to a bid just above zero. The rent K must be large enough to make that deviation unattractive, and in the worst case it is exactly large enough.

Equation (7) describes the bidding rule above the threshold. The common bid at value v is the expected value conditional on being in the bidding region, shaded by the probability that a win comes from the bidding region rather than from the pooling region. The comparison with the limiting BCE bidding rule $G(v) = \mathbb{E}[z \mid z \leq v]$ shows the source of the gap:

$$G(v) - H(v) = \frac{K}{F(v)} \quad \text{for } v \geq \hat{v}.$$

The coarse bidding rule equals the interim rule minus $K/F(v)$, and the difference between the two guarantees is the integral of this term against f .

Example 2 (Uniform values). Let $\mu = U[0, 1]$, so $m = 1/2$ and $F(v) = v$. Then $H(v) = (v^2 - \hat{v}^2)/(2v)$, and the system (5)–(6) reduces to the single equation

$$\frac{1}{2} - \frac{N\hat{v}^2}{2} = \frac{1}{4} - \frac{\hat{v}^2}{4} + \frac{\hat{v}^2 \log \hat{v}}{2}.$$

With two buyers the threshold is $\hat{v} \approx 0.6730$ and $CRG \approx 0.0471$. The seller collects less than 5% of the good's expected value. Since $F(\hat{v}) = \hat{v}$ for the uniform distribution, the pooling region has probability 0.673: in the worst case, both buyers bid zero with probability 0.673, and the good is then sold at a price of zero. Conditional on a

positive price being paid, the average price is 0.144. With ten buyers, $CRG \approx 0.1916$ and the pooling probability falls to 0.248.

The pooling probabilities in Example 2 motivate the reserve-price analysis of Section 6.

Proposition 2 (Comparative statics in N). *$CRG(\mu)$ is strictly increasing in N , and the rent K is strictly decreasing in N .*

The guarantee increases in N for a reason specific to coarse rationality. The rent that deters a fixed deviation is a per-buyer quantity: each buyer receives K , and by (3) the total is $NK = m - R$. With more buyers, each one's share of a given amount of uncollected surplus is smaller, so a given level of revenue leaves each buyer with a smaller equilibrium payoff, and the revenue reduction needed to satisfy the constraints is smaller.

5.3 Asymptotic Convergence and Its Rate

Theorem 2. *Under Assumption 1,*

$$\lim_{N \rightarrow \infty} CRG(\mu) = \int_0^{\bar{v}} G(v) f(v) dv = \lim_{N \rightarrow \infty} RG(\mu), \quad G(v) = \mathbb{E}[z \mid z \leq v].$$

The proof is in Appendix C. Intuitively, as N grows, the rent K vanishes at rate $1/N$, the pooling threshold $\hat{v}$ falls to zero, and by the identity $G - H = K/F$ the coarse bidding rule converges pointwise to the interim rule characterized by Bergemann, Brooks and Morris (2017).

A second argument for the same convergence connects the result to the information structure of the benchmark. The worst case in Bergemann, Brooks and Morris is supported by an information structure in which each bidder learns only whether they hold the highest value together with a summary of the remaining values. As N grows, that information structure converges to the value distribution itself, so conditioning on it adds no information, and the two notions of obedience impose the same restriction.

The next question is the rate of convergence. For pure common values, Bergemann, Brooks and Morris (2015), the working paper version of Bergemann, Brooks and Morris (2017), characterize the minimum-revenue BCE: the worst case has a

deterministic winning bid and the corresponding revenue

$$\beta_N(v) = \frac{\int_0^v z dF^{\frac{N-1}{N}}(z)}{F^{\frac{N-1}{N}}(v)}, \quad RG_N(\mu) = \int_0^{\bar{v}} \beta_N(v) f(v) dv, \quad (12)$$

the conditional mean of the value under the tilted distribution $F^{(N-1)/N}$.³ For the uniform prior, $\beta_N(v) = \frac{N-1}{2N-1}v$ and $RG_N = \frac{N-1}{2(2N-1)}$, which equals $1/6$ at $N = 2$ and increases to $1/4$. Based on this formula, the interim guarantee converges to the common limit at rate $1/N$. My next result shows that the coarse guarantee converges at a slower rate $\log N/N$.

Proposition 3 (Rate of convergence). *Let Assumption 1 hold and let R_∞ denote the limit in Theorem 2. Then $0 \leq R_\infty - RG_N \leq 2\bar{v}/(N-1)$, while*

$$R_\infty - CRG_N = \frac{(m - R_\infty) \log N}{2N} (1 + o(1)).$$

Consequently the cost of coarseness satisfies

$$RG_N - CRG_N = \frac{(m - R_\infty) \log N}{2N} (1 + o(1)) :$$

the interim guarantee converges at rate $1/N$, the coarse guarantee at rate $\log N/N$, and the gap between them closes at $\log N/N$.

The proof is in Appendix C. The gap $R_\infty - CRG_N$ equals the rent K times $\log(1/q_K)$, up to a term of order $1/N$, and K is of order $1/N$ for every prior. The rate therefore depends on how fast the pooling quantile q_K goes to zero, and this is the one place where the continuity and positivity of f at 0 in Assumption 1 are used. Near zero, $\Phi_\mu(q) \approx q^2/2f(0)$, hence $q_K \approx \sqrt{2f(0)K}$, and the square root is the factor $\frac{1}{2}$ in the rate. If instead f behaves like $x^{\alpha-1}$ near zero for some $\alpha > 0$, so that $f(0)$ is zero or infinite, then $q_K \approx K^{\alpha/(\alpha+1)}$, the rate is still $\log N/N$, and the factor $\frac{1}{2}$ is replaced by $\alpha/(\alpha+1)$.

Table 1 and Figure 1 illustrate these rates for the uniform prior, reporting the values at matched market sizes. With two bidders the coarse guarantee is under

³See Bergemann, Brooks and Morris (2015) for the common value case. In the as-if reading, choose the marginal $F^{1/N}$ so that the highest of N independent values is distributed as F , and then β_N is the textbook symmetric equilibrium bid of that model, the expected highest rival value conditional on winning, and the rival maximum has a distribution $F^{(N-1)/N}$.

N	2	3	5	10	20	50	100
Interim guarantee RG_N	0.1667	0.2000	0.2222	0.2368	0.2436	0.2475	0.2487
Coarse guarantee CRG_N	0.0471	0.0939	0.1453	0.1916	0.2178	0.2353	0.2419
Cost of coarseness $RG_N - CRG_N$	0.1196	0.1061	0.0770	0.0452	0.0258	0.0122	0.0068
CRG_N below RG_N (%)	71.7	53.1	34.6	19.1	10.6	4.9	2.7
Pooling probability $F(\hat{v})$	0.673	0.520	0.377	0.248	0.168	0.103	0.072

Table 1: Interim and coarse revenue guarantees for $\mu = U[0, 1]$, both converging to $1/4$. The interim row is $RG_N = \frac{N-1}{2(2N-1)}$, computed from the [Bergemann, Brooks and Morris \(2017\)](#) characterization via (12). The comparison is at the same market size throughout.

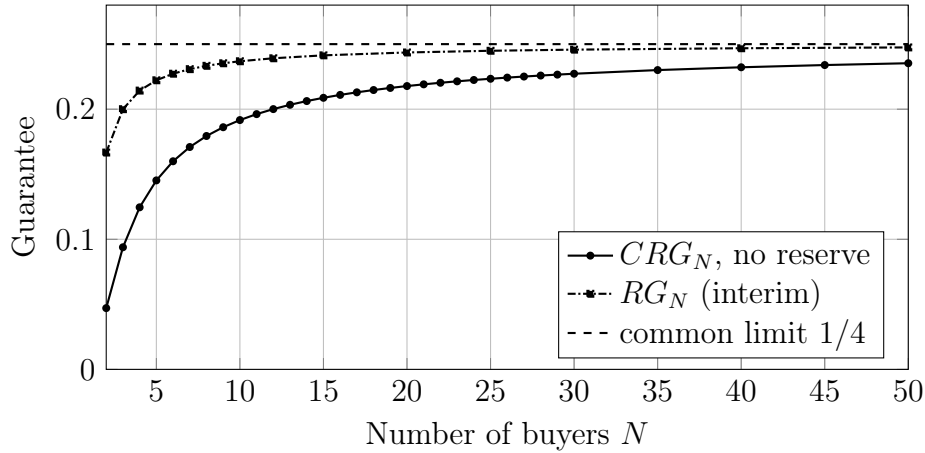


Figure 1: Interim and coarse revenue guarantees for $\mu = U[0, 1]$

thirty percent of the interim guarantee, and with ten it is about eighty percent.

The strict inequality between the RG and CRG is not specific to the uniform prior. The following result shows it holds for any μ under a finite number of buyers.

Proposition 4 (Strict coarseness gap). *Let Assumption 1 hold and suppose the infimum defining $RG_N(\mu)$ is attained by a BCE in which the two highest bids differ with positive probability, as in the construction of [Bergemann, Brooks and Morris \(2017\)](#). Then*

$$CRG_N(\mu) < RG_N(\mu) \quad \text{for every finite } N.$$

The proof, in Appendix B, uses only Proposition 1. In the minimum-revenue BCE of [Bergemann, Brooks and Morris](#), the winning bid is a deterministic function of the value and the losing bids lie strictly below it, so the two highest bids differ. Proposition 1 says that pooling such a distribution yields a BCCE with strictly lower

revenue. So the interim minimum is not a coarse minimum, and the two guarantees cannot be equal.

6 Reserve Prices

When N is small, the worst case of Sections 4 and 5 sells the good at a price of zero with high probability. This section shows that a reserve price raises the guarantee substantially if it is large enough. Example 2 shows the mechanism. The worst case is sustained by states in which every buyer bids zero and the buyers receive the good at a price of zero, and the resulting surplus is what satisfies their obedience constraints. A reserve price reduces this surplus. A small reserve reduces it only by the reserve itself, since bidders pool at the reserve and pay it, and only a reserve large enough that the worst case involves withholding eliminates it.

Formally, fix a reserve price $r \in [0, \bar{v})$. The first-price auction with reserve r allocates the good to the highest bidder at the bid price if the highest bid is at least r , breaking ties uniformly, and withholds the good otherwise. Write $CRG_N(r)$ for the resulting coarse revenue guarantee.

Bids below r never win, so without loss of generality every bid is either 0, interpreted as staying out, or at least r . The accounting identity (1) is amended in the following way, since the good is no longer always allocated:

$$\sum_i u_i(v, a) = (v - a_{(1)}) \mathbb{1}\{a_{(1)} \geq r\}. \quad (13)$$

The reduction holds with a modification. The pooling construction of Proposition 1 cannot be applied unchanged. Suppose exactly one buyer clears the reserve, so $a_{(1)} \geq r > a_{(2)}$. Pooling onto $a_{(2)}$ with probability $1/N$ would then withhold the good, and by (13) it would eliminate the buyers' surplus along with the seller's revenue, tightening rather than relaxing obedience. Instead, pool onto $\max\{a_{(2)}, r\}$, so that the good is still sold, and switch to withholding only in the states where the sale yields negative surplus, namely where the winning bid exceeds the value.

Proposition 5 (Identical Play with a Reserve). *Fix $r \in [0, \bar{v})$. For every BCCE σ of the first-price auction with reserve r there is an exchangeable identical play BCCE γ with $R(\gamma) \leq R(\sigma)$, where the common bid takes values in $\{0\} \cup [r, \bar{v}]$. Consequently $CRG_N(r)$ is the value of the identical play program.*

The detailed proof is in Appendix D.

Under a reserve, buyers can no longer receive the good at a price of zero, but they have a cheap alternative: pooling at exactly r , where the good still trades and the winner pays only the reserve. For values below r , however, buyers prefer not to buy at all. The worst case therefore involves up to two pools: a *withholding pool* at zero, where the good is not sold, and a *reserve pool* at exactly r , where the good trades at the reserve. The withholding pool occupies the lowest s quantiles, the reserve pool the next $q_0 - s$, and above q_0 the common bid is strictly increasing. The family is determined as follows. The cheapest deviation is a bid just above the reserve, which captures both pools outright, so obedience requires $K \geq \Phi_\mu(q_0) - r q_0$. And the bidding rule must start at the reserve, $(\Phi_\mu(q_0) - K)/q_0 \geq r$, which is the same inequality reversed. Hence at the worst case

$$K = \Phi_\mu(q_0) - r q_0. \quad (14)$$

Withholding removes value $\Phi_\mu(s)$, so

$$NK = m - \Phi_\mu(s) - R, \quad R = r(q_0 - s) + \int_{q_0}^1 \frac{\Phi_\mu(q) - K}{q} dq, \quad (15)$$

which, given q_0 , determines the withholding share $s \in [0, q_0]$. Let $\widehat{V}_N(r)$ denote the minimum of R over this one-parameter family, subject to $K \geq 0$ and the feasibility condition $\Phi_\mu(q_0) - \Phi_\mu(s) \geq r(q_0 - s)$, which says that the average value in the reserve pool is at least r .

Theorem 3 (Reserve prices). *Let $r \in [0, m)$ satisfy $rF(r) - \Phi_\mu(F(r)) < (N-1)(m-r)$, which is true for every $r \leq m(N-1)/N$ and for every r in the example below. Then*

$$CRG_N(r) = \widehat{V}_N(r),$$

so the two-pool family is the worst case. Moreover, the guarantee is continuous at a zero reserve, $\lim_{r \downarrow 0} CRG_N(r) = CRG_N(0)$.⁴ For $r \geq m$ the guarantee is zero, since no participation is a BCCE.

The proof is in Appendix D and uses two steps beyond those of Theorem 1. The

⁴The characterization was also confirmed by directly solving a discretized version of the coarse program as a linear program, with neither identical play nor comonotonicity imposed. The linear program reproduces $\widehat{V}_N(r)$ at every reserve tested.

r	0.01	0.05	0.10	0.148	0.20
$CRG_2(r)$	0.0552	0.0812	0.1000	0.1056	0.1038
withholding share s	0.051	0.253	0.464	0.621	0.664
pool boundary q_0	0.676	0.674	0.650	0.621	0.664

Table 2: The worst case across reserve levels, $\mu = U[0, 1]$, $N = 2$. The reserve pool has mass $q_0 - s$. It contains most of the pooled mass at small reserves, shrinks as the reserve grows, and vanishes at the revenue-maximizing reserve, $r \approx 0.148$, beyond which the worst case relies on withholding alone. Values computed from (14)–(15).

first is a comonotonization lemma, which plays the role at the level of couplings that the pooling kernel plays at the level of bid profiles in Proposition 1. Rearranging the bid-value coupling into its comonotone form, holding the bid marginal fixed, leaves revenue unchanged, raises the buyers’ rent, and lowers every deviation payoff, so the worst case couples bids and values comonotonically. The second part reduces the remaining problem to a system of equations similar to 1, and argues that the two-pool family identical play attains the lower bound.

Example 3 (Uniform values with a reserve). Return to the setting of Example 2, where $\mu = U[0, 1]$, $N = 2$, and without a reserve $CRG_2 \approx 0.047$, sustained by both buyers bidding zero on the lowest two-thirds of values and receiving the good at a price of zero. Table 2 traces the worst case as the reserve grows. At $r = 0.01$ nearly the entire former pooling region bids r , almost nothing is withheld, and the guarantee rises continuously from $CRG_2(0)$, with slope roughly 0.8. As the reserve grows, pooling at the reserve becomes costly for the buyers and the worst case shifts toward withholding, so the share s rises. The guarantee is maximized at $r \approx 0.148$, which is the point where the reserve pool vanishes. At a smaller reserve part of the pooling region still bids r , and at a larger one the probability of sale falls without further tightening obedience. At that reserve $CRG_2 \approx 0.106$, more than double the guarantee without one, and the gain accrues gradually, since at smaller reserves part of the pooling region bids r : reaching half of it requires $r \approx 0.041$. With two coarse bidders, the guarantee without a reserve is less than half of the guarantee at the optimal reserve.

7 Beyond Common Values: Symmetric Priors

The analysis so far, with and without a reserve, fixes the pure common value model. Since the interim characterization of [Bergemann, Brooks and Morris \(2017\)](#) covers symmetric priors, this section asks how much of the analysis extends to values that may differ across buyers. Let π be a prior on V^N that is exchangeable, that is, invariant under every permutation of the coordinates, with common per-buyer marginal μ , let buyer i 's payoff be $q_i(a)(v_i - a_i)$, and define BCCE exactly as in Lemma 1, now over joint laws of $(v_1, \dots, v_N, a_1, \dots, a_N)$ whose marginal on the value profile is π . Such a law is exchangeable if it is invariant under permuting values and bids together, $(v, a) \mapsto (v_\rho, a_\rho)$. Pure common values are the coupling $v_1 = \dots = v_N$, and independent private values are the opposite extreme. Write $CRG(\pi)$ for the infimum of revenue over all BCCE under π . Under common values, Proposition 1 reduces this infimum to identical play. This section asks whether the reduction holds when values differ, and what the guarantee is when it does not.

The reason the coupling matters is specific to coarse obedience. Under BCE, the correlation structure of the prior is irrelevant to the seller's problem, in the sense that the method of [Bergemann, Brooks and Morris \(2017\)](#) handles every coupling. Under BCCE it matters. The contrast comes from how each solution concept constrains a bidder. Under BCE, obedience is checked conditional on the bidder's own recommendation, so nature can correlate each bidder's signal with that bidder's own value. This gives nature N separate instruments, one per bidder, and it is why the [Bergemann, Brooks and Morris](#) worst case handles arbitrary couplings and is even allocatively efficient. Under BCCE, a bidder's own bid appears nowhere in their own deviation payoff $\mathbb{E}[(v_i - \delta) \mathbb{1}\{\max_{j \neq i} a_j < \delta\}]$, so the only thing that constrains buyer i is the correlation between v_i and the *other* buyers' bids.

Turning to the reduction, whether the identical play program gives the guarantee away from common values depends on whether the reduction of Proposition 1 holds, and it does not in general. The symmetrization half still holds. It is still possible to make every buyer symmetric without changing the revenue (Lemma 13). The pooling half, however, fails. The pooling kernel of Lemma 4 allocates the good to a uniformly random buyer instead of to the top bidder. Under common values this changes nothing, since every buyer values the good equally. When values differ, it changes two things. If the top bidder has above-average value, the buyers' total surplus falls

when the good is reassigned at random, so the pooled rent can fall rather than rise. And a deviator who outbids the others but not the top bidder is, by exchangeability, someone other than the top bidder, so the deviation payoff is no longer unchanged. Both corrections have the same sign: pooling still works when $\hat{\sigma}$ allocates against value and fails when it allocates efficiently. With heterogeneous values the adversary lowers revenue by allocating efficiently: the good goes to the buyer with the highest value, the buyers' total surplus rises, and the additional surplus relaxes the obedience constraints, which permits lower bids.

Moreover, in the first-price auction the adversary can do this at an arbitrarily small cost in revenue. Fix a common price and give the buyer with the highest value a bid premium of ε above it. The premium decides the allocation and adds only ε to revenue, while every coarse deviation still faces, up to ε , the common price as its highest competing bid. The next result shows that this is the generalization of Theorem 1. Let $m_N := \mathbb{E}_\pi[\max_i v_i]$ denote the expected highest value, so that $m \leq m_N \leq Nm$, and let $\tilde{K}$ be the unique solution in $(0, m]$ of

$$m_N - NK = \mathcal{R}_\mu(K), \quad (16)$$

which exists by the argument of Lemma 8 with m_N in place of m . Write $\tilde{q} := \Phi_\mu^{-1}(\tilde{K})$ and $\tilde{v} := F^{-1}(\tilde{q})$ for the associated pooling quantile and threshold.

Theorem 4 (Symmetric priors). *Let π be exchangeable with per-buyer marginal μ satisfying Assumption 1. Then*

$$CRG(\pi) \geq m_N - N\tilde{K} = \mathcal{R}_\mu(\tilde{K}).$$

The inequality holds with equality if $v_1 = \dots = v_N$ holds π -almost surely on $\{\max_i v_i > \tilde{v}\}$ and, with positive probability, at least two buyers have strictly positive values on $\{\max_i v_i \leq \tilde{v}\}$.

The proof is in Appendix E. The lower bound is the chain of Theorem 1 with two substitutions: the rearrangement inequality is applied to the pair (v_i, M_i) of a buyer's own value and highest competing bid, which uses only the marginal μ , and the accounting identity bounds the buyers' total surplus by the efficient surplus $m_N - R$ rather than by $m - R$. The attainment construction was described above. On the pooling event $\{\max_i v_i \leq \tilde{v}\}$ the highest-value buyer bids ε and the others bid zero,

above it all buyers bid a common function of the common value, and as $\varepsilon \downarrow 0$ revenue converges to the bound. The non-degeneracy condition in the equality case rules out the case in which at most one buyer ever has a positive value in the pool, where the premium cannot raise the buyers' total surplus.

One corollary is that, in the equality case, the guarantee lies strictly below the common value guarantee whenever the values differ with positive probability on the pooling event, since then $m_N > m$, the fixed point (16) lies above the fixed point (10), and $\mathcal{R}_\mu$ is strictly decreasing:

$$CRG(\pi) = \mathcal{R}_\mu(\tilde{K}) < \mathcal{R}_\mu(K^*) = CRG(\mu).$$

8 Ranking and Equivalence Across Auctions

I now return to common values and compare the first-price auction with other formats, following Bergemann, Brooks and Morris (2019). Following their framework, a *standard auction* is one that allocates the good to the highest bidder, breaking ties uniformly, and admits an equilibrium in monotonic pure strategies in the symmetric independent private values model. Losers pay nothing, and the winner pays $p(a)$, which I take to satisfy $p(a) \leq a_{(1)}$ and $p(b, \dots, b) = b$: the winner never pays more than their bid, and tied bidders pay their common bid. Let $\mathcal{M}$ denote this class. Within it, the pricing rules that matter are defined by what happens on profiles in which the $N - 1$ losing bids coincide. A standard auction has *own-bid pricing* if the winner pays $a_{(1)}$ whenever the other bids coincide, as in the first-price and Dutch auctions, and *competing-bid pricing* if the winner pays $a_{(2)}$ whenever the other bids coincide, as in the second-price and English auctions.

The comparison uses the following two facts, combined as in Bergemann, Brooks and Morris. The first is that identical play is an outcome every standard auction can produce, at the same revenue. The second is that formats differ only in what a single deviator pays after outbidding the others.

Proposition 6 (Identical play is format-free). *Fix any identical play distribution with common bid c . In every standard auction the good is allocated uniformly among the N buyers, the winner pays c , and expected revenue is $\mathbb{E}[c]$. In every standard auction with own-bid pricing, moreover, the identical play BCCE are exactly those*

of the first-price auction, so the coarse revenue guarantee restricted to identical play equals $CRG(\text{FPA})$ in each of them.

Proof. Under own-bid pricing, a buyer who deviates from an identical play distribution to the fixed bid δ loses if $\delta < c$, ties at price c if $\delta = c$, and wins at price δ if $\delta > c$, exactly as in the first-price auction, and equilibrium payoffs coincide by the first part. So the coarse constraints (2) are the same in both formats, and the sets of identical play BCCE coincide. $\square$

Proposition 7 (Coarse revenue guarantee ranking). *The first-price auction has the highest coarse revenue guarantee among standard auctions with own-bid or competing-bid pricing. For every $M \in \mathcal{M}$ with own-bid pricing, $CRG(M) \leq CRG(\text{FPA})$, and for every $M \in \mathcal{M}$ with competing-bid pricing, in particular the second-price and English auctions,*

$$CRG(M) = 0 < CRG(\text{FPA}).$$

Proof. Under own-bid pricing, the identical play BCCE attaining $CRG(\text{FPA})$ in Theorem 1 is a BCCE of M with the same revenue, by Proposition 6, so $CRG(M) \leq CRG(\text{FPA})$. Under competing-bid pricing, consider the distribution in which one designated buyer bids $\bar{v}$ and all others bid zero, independently of the value. Revenue is zero. The designated buyer earns m and cannot improve, since any positive bid still wins at the price $p(\delta, 0, \dots, 0) = 0$. Any other buyer must bid $\bar{v}$ to contest the good, which ties the top bid and, by $p(b, \dots, b) = b$, means winning with some probability $\lambda \in (0, 1]$ at price $\bar{v}$, for a payoff of $\lambda(m - \bar{v}) \leq 0$, and the conclusion does not depend on how ties are broken. This is a BCCE with zero revenue, so $CRG(M) = 0$, and $CRG(\text{FPA}) > 0$ by Theorem 1. $\square$

9 Unknown Value Distribution

Up to here, it is assumed that the seller knows the prior distribution. What would happen if the seller knows only the expected value $E = \mathbb{E}[v]$ and the upper bound $\bar{v}$ on the value, but not the shape of the distribution? This section extends the analysis to an unknown value distribution. The worst case now has a binary distribution, and that violates the density assumption of Assumption 1, and I now define everything in the non-density format, although with heavier notation.

Theorem 5 (Beyond the density). *Let μ be any non-degenerate distribution on $[0, \bar{v}]$, and read the definitions (8), (9), and (11) with the quantile function $F^{-1}(t) = \inf\{x : F(x) \geq t\}$ in place of the inverse. Then (10) has a unique solution $K^* \in (0, m)$, and $CRG(\mu) = m - NK^* = \mathcal{R}_\mu(K^*) > 0$. The infimum is attained by the identical play distribution $v = F^{-1}(U)$, $c = \psi(U)$, with U uniform on $[0, 1]$.*

In the attaining distribution, the uniform variable is the rank of the value. Off atoms $U = F(v)$, so the coupling reduces to the deterministic bid $H(v) = \psi(F(v))$ of Theorem 1. Conditional on an atom of μ , however, U is uniform on the corresponding quantile interval, so the common bid is randomized continuously across it. The randomization is required by the constraints. Minimality requires the obedience constraint to bind at every deviation level, and a deviation to $\psi(q)$ with q inside an atom's interval must capture exactly a fraction of the atom, which a single common bid cannot arrange. If the whole atom bid one number, a deviation just above it would capture the entire mass at once and earn strictly more than the rent.

The proof, in Appendix F, checks that the argument of Appendix B holds with atoms. The identical play reduction of Proposition 1 does not use the density. Three steps of the characterization use it, and each is modified: the rearrangement inequality holds for arbitrary priors once its equality clause allows an atom to be split, the concavity lemma behind uniqueness does not use the density, and the attaining coupling randomizes the common bid at any atom of the prior, in the manner just described.

With the characterization extended to atomic priors, the relevant comparison across priors is the convex order: $\mu \preceq_{\text{cx}} \nu$, read as “ ν is riskier than μ ,” means that μ and ν have the same mean and that ν can be obtained from μ by a sequence of mean-preserving spreads.

Proposition 8 (Convex order monotonicity). *If $\mu \preceq_{\text{cx}} \nu$ then $CRG(\nu) \leq CRG(\mu)$, for every N . Consequently, for every N ,*

$$\min_{\mu: \text{supp } \mu \subseteq [0, \bar{v}], \mathbb{E}_\mu[v] = E} CRG(\mu) = CRG(\underline{\mu}),$$

where $\underline{\mu}$ is the binary distribution with $\underline{\mu}(\{0\}) = 1 - E/\bar{v}$ and $\underline{\mu}(\{\bar{v}\}) = E/\bar{v}$.

The proof is in Appendix F. What sustains a low price in the worst case is that a buyer who deviates to a slightly higher bid wins mostly in states where the value is low, so the deviation is unprofitable despite the low price. This relies on the bid being

increasing in the value. A mean-preserving spread moves probability mass toward the extremes and weakens this relation, so a larger rent is needed to satisfy the obedience constraints, and revenue falls. Since the binary distribution on $\{0, \bar{v}\}$ is the riskiest distribution on $[0, \bar{v}]$ with a given mean, it is the seller's worst case.

Corollary 1 (The binary guarantee). *Let $p_0 = 1 - E/\bar{v}$. For finite N , $CRG(\underline{\mu}) = E - N\bar{v}(q^* - p_0)$, where $q^* \in (p_0, 1)$ is the unique solution of $(N - 1)(p_0 - q) = q \log q$. As $N \rightarrow \infty$,*

$$CRG(\underline{\mu}) \rightarrow E + (\bar{v} - E) \log \left(1 - \frac{E}{\bar{v}}\right). \quad (17)$$

Stating the result through the convex order, rather than through the extreme point alone, has three advantages. It holds at every N , not only in the limit. It covers distributions with atoms, which matters because the minimizer is itself atomic. And it gives a full comparative static rather than just the worst case, so a seller with partial information about dispersion can locate their guarantee between the extremes. For $E = 1/2$ and $\bar{v} = 1$, the ordering $U[0, 1] \preceq_{\text{cx}} \underline{\mu}$ corresponds to the reverse ordering of guarantees: 0.047 against 0.040 at $N = 2$, and 0.250 against 0.153 in the limit. A more dispersed prior lowers the guarantee.

10 Conclusion

This paper characterizes the worst-case revenue of auctions when bidders are rational only in the coarse sense that no single fixed bid would have served them better on average. No-regret learning leads to this restriction, and it is the restriction that best describes algorithmic bidders. In the first-price auction the worst case is an identical play equilibrium, in which bidders do not use their private information and all submit the same bid, and this reduction yields a closed-form characterization of the guarantee. The guarantee is strictly positive, increasing in the number of bidders, and strictly below the interim benchmark of [Bergemann, Brooks and Morris \(2017\)](#) at every finite market size. Both converge to a common limit, but the interim guarantee converges at rate $1/N$ while the coarse guarantee at rate $\log N/N$. With a symmetric prior, the identical play is no longer a worst case, but the coarse revenue guarantee is still positive and lower bounded by a similar system of equations to the common value case.

The worst case is sustained by states in which every bidder bids zero and the buyers receive the good at a price of zero, and the resulting surplus is what satisfies their obedience constraints. A reserve price reduces this surplus, but a small reserve reduces it only slightly: coarse bidders pool at the reserve and pay it, the surplus falls only by the reserve, and the guarantee is continuous at a zero reserve. The revenue-maximizing reserve is the smallest reserve at which the reserve pool vanishes and the worst case involves withholding only. Across formats, the first-price auction has the highest guarantee: identical play is available at the same revenue in every standard auction, and where a deviator pays the others' bid rather than their own, the guarantee is zero. Finally, the guarantee is monotone in the convex order, so a seller who knows only the mean should benchmark against the binary distribution.

For a seller who faces learning algorithms, the results suggest the following. Use the first-price auction, since it has the highest guarantee. Set a reserve price large enough that withholding, not pooling at the reserve, is the worst case, since a small reserve is paid by the buyers and changes little. When the value distribution is unknown, use the binary distribution with the same mean as the benchmark.

Several questions remain. Away from common values the guarantee is characterized exactly only for priors that agree on the bidding region. For priors that do not satisfy this condition, the coarse revenue guarantee remains open. Whether the first-price auction maximizes the coarse revenue guarantee over all standard auctions, as it maximizes the interim guarantee in [Bergemann, Brooks and Morris \(2019\)](#), is also open, since the ranking proved here covers the formats in which a deviator who outbids tied opponents pays either their own bid or the opponents' bid. And the optimal mechanism against coarsely rational bidders, of which the reserve price studied here is only one simple instrument, remains open as well. Finally, [Remark 1](#) shows that the guarantee for contextual algorithms lies between the coarse and the interim guarantee, and characterizing it is left for future work.

References

- Azacis, Helmut, and Peter Vida.** 2015. “Collusive communication schemes in a first-price auction.” *Economic Theory*, 58(1): 125–160.
- Battigalli, Pierpaolo, and Marciano Siniscalchi.** 2003. “Rationalizable bidding in first-price auctions.” *Games and Economic Behavior*, 45(1): 38–72. First World Congress of the Game Theory Society.
- Bergemann, Dirk, and Stephen Morris.** 2016. “Bayes Correlated Equilibrium and the Comparison of Information Structures in Games.” *Theoretical Economics*, 11(2): 487–522.
- Bergemann, Dirk, Benjamin Brooks, and Stephen Morris.** 2015. “First Price Auctions with General Information Structures: Implications for Bidding and Revenue.” Cowles Foundation for Research in Economics, Yale University Cowles Foundation Discussion Paper 2018.
- Bergemann, Dirk, Benjamin Brooks, and Stephen Morris.** 2017. “First-Price Auctions With General Information Structures: Implications for Bidding and Revenue.” *Econometrica*, 85(1): 107–143.
- Bergemann, Dirk, Benjamin Brooks, and Stephen Morris.** 2019. “Revenue Guarantee Equivalence.” *American Economic Review*, 109(5): 1911–29.
- Blum, Avrim, MohammadTaghi Hajiaghayi, Katrina Ligett, and Aaron Roth.** 2008. “Regret minimization and the price of total anarchy.” *Proceedings of the Annual ACM Symposium on Theory of Computing*, 373–382.
- Braverman, Mark, Jieming Mao, Jon Schneider, and S. Matthew Weinberg.** 2018. “Selling to a No-Regret Buyer.” 523–538.
- Brooks, Benjamin, and Songzi Du.** 2021. “Optimal Auction Design With Common Values: An Informationally Robust Approach.” *Econometrica*, 89(3): 1313–1360.
- Bulow, Jeremy, and Paul Klemperer.** 1996. “Auctions Versus Negotiations.” *American Economic Review*, 86(1): 180–194.

- Dekel, Eddie, and Asher Wolinsky.** 2003. “Rationalizable outcomes of large private-value first-price discrete auctions.” *Games and Economic Behavior*, 43(2): 175–188.
- Deng, Yuan, Jon Schneider, and Balasubramanian Sivan.** 2019. “Strategizing against No-Regret Learners.”
- Du, Songzi.** 2018. “Robust Mechanisms Under Common Valuation.” *Econometrica*, 86(5): 1569–1588.
- Fang, Hanming, and Stephen Morris.** 2006. “Multidimensional private value auctions.” *Journal of Economic Theory*, 126(1): 1–30.
- Feldman, Michal, Brendan Lucier, and Noam Nisan.** 2016. “Correlated and Coarse Equilibria of Single-Item Auctions.” 131–144. Berlin, Heidelberg:Springer Berlin Heidelberg.
- Hardy, G. H., J. E. Littlewood, and G. Pólya.** 1934. *Inequalities*. Cambridge:Cambridge University Press.
- Hartline, Jason D, Vasilis Syrgkanis, and Éva Tardos.** 2015. “No-regret learning in Bayesian games.” *Advances in Neural Information Processing Systems*, 2015-January: 3061–3069.
- Hart, Sergiu, and Andreu Mas-Colell.** 2000. “A Simple Adaptive Procedure Leading to Correlated Equilibrium.” *Econometrica*, 68(5): 1127–1150.
- Jin, Yaonan, and Pinyan Lu.** 2023. “First Price Auction is $1-1/e^2$ Efficient.” *J. ACM*, 70(5).
- Kolumbus, Yoav, and Noam Nisan.** 2022. “Auctions between Regret-Minimizing Agents.” 100–111.
- Lomys, Niccolò, and Lorenzo Magnolfi.** 2025. “Estimation of Games under No Regret: Structural Econometrics for AI.” *Working Paper*.
- Milgrom, Paul R., and Robert J. Weber.** 1982. “A Theory of Auctions and Competitive Bidding.” *Econometrica*, 50(5): 1089–1122.

- Moulin, Hervé, and Jean-Philippe Vial.** 1978. “Strategically zero-sum games: The class of games whose completely mixed equilibria cannot be improved upon.” *International Journal of Game Theory*, 7: 201–221.
- Nekipelov, Denis, Vasilis Syrgkanis, and Eva Tardos.** 2015. “Econometrics for Learning Agents.” 1–18.
- Shaked, Moshe, and J. George Shanthikumar.** 2007. *Stochastic Orders*. New York:Springer.
- Syrgkanis, Vasilis, and Eva Tardos.** 2013. “Composable and efficient mechanisms.” *STOC ’13*, 211–220. New York, NY, USA:Association for Computing Machinery.
- Vickrey, William.** 1961. “Counterspeculation, Auctions, And Competitive Sealed Tenders.” *The Journal of Finance*, 16(1): 8–37.

A Proofs for Sections 3 and 4

Proof of Lemma 2

Fix δ and write $A := \mathbb{E}[(v - \delta)\mathbb{1}\{c < \delta\}]$ and $B := \mathbb{E}[(v - \delta)\mathbb{1}\{c \leq \delta\}]$. A deviator bidding δ wins outright when $c < \delta$ and shares with the $N - 1$ other buyers when $c = \delta$, so the deviation payoff is $(1 - \frac{1}{N})A + \frac{1}{N}B \leq \max\{A, B\}$. Taking $\delta_n \uparrow \delta$ and $\delta_n \downarrow \delta$ in D and applying dominated convergence to the hypothesis gives $K \geq A$ and $K \geq B$. At $\delta = 0$ only the downward limit is available, and it gives $K \geq B \geq 0 = A$. At $\delta = \bar{v}$ only the upward limit is available, and it gives $K \geq A$, while $B = \mathbb{E}[v - \bar{v}] \leq 0 \leq K$. $\square$

The identical play reduction

Throughout, for an action profile $a \in A$ let $a_{(1)} \geq a_{(2)} \geq \dots \geq a_{(N)}$ denote the order statistics. The map $a \mapsto (a_{(1)}, a_{(2)})$ is continuous and hence measurable. For $c \in [0, \bar{v}]$ write $\mathbf{c} = (c, \dots, c)$.

Lemma 3 (Symmetrization). *For $\sigma \in \mathcal{BCC}\mathcal{E}(\mu)$ let $\hat{\sigma} := \frac{1}{N!} \sum_{\rho} \sigma^{\rho}$, where σ^{ρ} is the image of σ under $(v, a) \mapsto (v, a_{\rho})$ with $(a_{\rho})_i = a_{\rho(i)}$. Then $\hat{\sigma} \in \mathcal{BCC}\mathcal{E}(\mu)$ is exchangeable and $R(\hat{\sigma}) = R(\sigma)$.*

Proof. Permuting action coordinates leaves the value marginal equal to μ , since the value is common. Since $u_i(v, a_{\rho}) = u_{\rho(i)}(v, a)$ and a coarse deviation is a constant, buyer i 's constraint under σ^{ρ} is buyer $\rho(i)$'s constraint under σ , so each σ^{ρ} lies in $\mathcal{BCC}\mathcal{E}(\mu)$. The constraints (2) are linear in σ , so $\mathcal{BCC}\mathcal{E}(\mu)$ is convex and the average $\hat{\sigma}$ belongs to it. Finally $\max_i a_i$ is permutation invariant and R is linear, so revenue is unchanged. $\square$

Lemma 4 (Pooling). *Let $\hat{\sigma} \in \mathcal{BCC}\mathcal{E}(\mu)$ be exchangeable and let γ be the image of $\hat{\sigma}$ under the Markov kernel*

$$\Gamma((v, a), \cdot) = \delta_{\{v\}} \otimes \left[\frac{N-1}{N} \delta_{\{\mathbf{a}_{(1)}\}} + \frac{1}{N} \delta_{\{\mathbf{a}_{(2)}\}} \right].$$

Then $\gamma \in \mathcal{BCC}\mathcal{E}(\mu)$ is exchangeable identical play, and

$$R(\gamma) = \mathbb{E}_{\hat{\sigma}} \left[\frac{N-1}{N} a_{(1)} + \frac{1}{N} a_{(2)} \right] \leq R(\hat{\sigma}),$$

strictly unless $a_{(1)} = a_{(2)}$ holds $\widehat{\sigma}$ -almost surely.

Proof. It is obvious that γ is identical play, exchangeable, and has value marginal μ and the revenue under γ is the common bid. Since $a_{(2)} \leq a_{(1)}$ and $R(\widehat{\sigma}) = \mathbb{E}_{\widehat{\sigma}}[a_{(1)}]$, revenue weakly falls, where equality is attained only if the top two bids are equal almost surely.

Step 1: the law of the highest competing bid is unchanged. Fix a buyer i and let $M_i := \max_{j \neq i} a_j$. Condition on (v, a) under $\widehat{\sigma}$ and average across the N buyers, which exchangeability permits. If $a_{(1)} > a_{(2)}$, exactly one buyer, the unique top bidder, faces $M_i = a_{(2)}$, and the remaining $N - 1$ face $M_i = a_{(1)}$. If $a_{(1)} = a_{(2)}$, every buyer faces $M_i = a_{(1)}$. In both cases the conditional law of M_i is $\frac{N-1}{N}\delta_{\{a_{(1)}\}} + \frac{1}{N}\delta_{\{a_{(2)}\}}$, which by construction of Γ is exactly the conditional law of the common bid c under γ . Hence (v, M_i) and (v, c) have the same joint law.

Step 2: deviation payoffs. Let D be the set of δ that are not atoms of the law of c . Its complement is countable, so D is dense. Fix $\delta \in D$. By Step 1, δ is not an atom of the law of M_i either, so ties occur with probability zero under both distributions and

$$\mathbb{E}_{\gamma}[(v - \delta)\mathbb{1}\{c < \delta\}] = \mathbb{E}_{\widehat{\sigma}}[(v - \delta)\mathbb{1}\{M_i < \delta\}] = (\text{buyer } i\text{'s deviation payoff under } \widehat{\sigma}).$$

Step 3: equilibrium payoffs weakly rise. The good is always allocated, so by (1) the buyers' total payoff is $v - a_{(1)}$ under $\widehat{\sigma}$ and $v - c$ under γ . Both distributions are exchangeable, so each buyer's expected payoff is $1/N$ of the expected total:

$$\mathbb{E}_{\widehat{\sigma}}[u_i] = \frac{1}{N}\mathbb{E}_{\widehat{\sigma}}[v - a_{(1)}] \leq \frac{1}{N}\mathbb{E}_{\widehat{\sigma}}\left[v - \frac{N-1}{N}a_{(1)} - \frac{1}{N}a_{(2)}\right] = \mathbb{E}_{\gamma}[u_i] =: K_{\gamma}.$$

Combining, for every $\delta \in D$ we have $K_{\gamma} \geq \mathbb{E}_{\widehat{\sigma}}[u_i] \geq \mathbb{E}_{\widehat{\sigma}}[(v - \delta)\mathbb{1}\{M_i < \delta\}] = \mathbb{E}_{\gamma}[(v - \delta)\mathbb{1}\{c < \delta\}]$. Lemma 2 extends the inequality to every $\delta \in [0, \bar{v}]$, ties included. Hence $\gamma \in \mathcal{BCC}\mathcal{E}(\mu)$. $\square$

Proof of Proposition 1. Compose Lemmas 3 and 4. The statement is a map from an arbitrary BCCE to an identical play BCCE with weakly lower revenue, so no minimizer need be presumed to exist, and the equality of the two infima follows. $\square$

B Proof of Theorem 1

Assumption 1 is in force, so F is continuous and strictly increasing on $[0, \bar{v}]$ with inverse F^{-1} , and Φ_μ , $\mathcal{R}_\mu$, and ψ are as in (8), (9), and (11). I prove the theorem in its fixed-point form: the equation (10) has a unique solution $K^* \in (0, m)$, no identical play BCCE, and hence by Proposition 1 no BCCE at all, yields revenue below $m - NK^*$, and the comonotone coupling $a = \psi(F(v))$ attains it, so that $CRG(\mu) = m - NK^* = \mathcal{R}_\mu(K^*) > 0$. Theorem 1 follows by the substitution $q = F(v)$: the threshold is defined through $\Phi_\mu(F(\hat{v})) = K^*$, that is, $\int_0^{\hat{v}} z f(z) dz = K^*$. Then $\psi(F(v)) = H(v)$ as in (7) and (10) becomes (5)–(6).

Throughout, $v \sim \mu$, $R = \mathbb{E}[c]$, $K = (m - R)/N$, and $A(q) = \inf\{x : \Pr(c \leq x) \geq q\}$ is the quantile function of the bid, so $R = \int_0^1 A(q) dq$.

Lemma 5 (Rearrangement bound). *For any event B with $\Pr(B) = q$, $\mathbb{E}[v \mathbb{1}_B] \geq \Phi_\mu(q)$, with equality if and only if B coincides with the lower-contour set $\{v \leq F^{-1}(q)\}$ up to null sets.*

Proof. This is the Hardy–Littlewood rearrangement inequality (Hardy, Littlewood and Pólya, 1934): among events of probability q , the expectation $\mathbb{E}[v \mathbb{1}_B]$ is smallest when B is placed on the lowest realizations of v , which gives $\int_0^q F^{-1}(t) dt$. $\square$

By Lemma 5, monotonicity of the worst-case bidding rule is derived rather than assumed. The next two lemmas are the two halves of the argument, a lower bound and an attainment check, in a form that is reused for reserve prices, symmetric priors, and atomic priors. Neither uses the density in Assumption 1: both hold for every prior once F^{-1} is read as the quantile function.

Lemma 6 (Chain). *Let X be a random variable with values in $[0, \bar{v}]$ on the same probability space as $v \sim \mu$, with cdf F_X and quantile function $A(q) = \inf\{x : F_X(x) \geq q\}$. Let $K \geq 0$ and $\Delta \subseteq [0, \bar{v}]$, and suppose*

$$K \geq \mathbb{E}[(v - \delta) \mathbb{1}\{X \leq \delta\}] \quad \text{for every } \delta \in \Delta.$$

Then $A(q) \geq (\Phi_\mu(q) - K)/q$ for every $q \in (0, 1)$ with $A(q) \in \Delta$. In particular, if $\Delta = [0, \bar{v}]$, then $\mathbb{E}[X] \geq \mathcal{R}_\mu(K)$.

Proof. By Lemma 5, the hypothesis gives $K \geq \Phi_\mu(F_X(\delta)) - \delta F_X(\delta)$ for every $\delta \in \Delta$. Fix q with $\delta = A(q) \in \Delta$. Right-continuity of F_X gives $F_X(A(q)) \geq q$, and the map

$p \mapsto (\Phi_\mu(p) - K)/p$ is nondecreasing, since its derivative has the sign of $F^{-1}(p)p - \Phi_\mu(p) + K = \int_0^p (F^{-1}(p) - F^{-1}(t))dt + K \geq 0$. Hence $A(q) \geq (\Phi_\mu(F_X(A(q))) - K)/F_X(A(q)) \geq (\Phi_\mu(q) - K)/q$. If $\Delta = [0, \bar{v}]$ this holds for every q , and since $A \geq 0$, $\mathbb{E}[X] = \int_0^1 A(q) dq \geq \int_0^1 (\Phi_\mu(q) - K)^+/q dq = \mathcal{R}_\mu(K)$. $\square$

Lemma 7 (Binding bids). *Let U be uniform on $[0, 1]$, $v = F^{-1}(U)$, and $c = C(U)$, where $C : [0, 1] \rightarrow [0, \bar{v}]$ is nondecreasing and, for some $q_0 \in [0, 1)$ and $K \geq 0$, satisfies $C(q) = (\Phi_\mu(q) - K)/q$ on $(q_0, 1]$. If $K > 0$ or Assumption 1 holds, then C is continuous and strictly increasing on $(q_0, 1]$, and the identical play distribution with common bid c satisfies*

$$\mathbb{E}\left[(v - \delta)\left(\mathbb{1}\{c < \delta\} + \frac{1}{N}\mathbb{1}\{c = \delta\}\right)\right] \leq K \quad \text{for every } \delta > C(q_0^+),$$

with equality for $\delta \in (C(q_0^+), C(1)]$, where $C(q_0^+) = \lim_{q \downarrow q_0} C(q)$.

Proof. On $(q_0, 1]$, Φ_μ is absolutely continuous and $C'(q) = (F^{-1}(q)q - \Phi_\mu(q) + K)/q^2 = (\int_0^q (F^{-1}(q) - F^{-1}(t))dt + K)/q^2$ almost everywhere. This is positive when $K > 0$, and also when $K = 0$ under Assumption 1, since F^{-1} is then strictly increasing. So C is continuous and strictly increasing on $(q_0, 1]$, with $C(1) = m - K$. Fix $\delta \in (C(q_0^+), C(1)]$ and let $q \in (q_0, 1]$ be the unique solution of $C(q) = \delta$. Since C is nondecreasing and $C(q_0^+) < \delta$, $\{c < \delta\} = \{U < q\}$ up to a null set and $\Pr(c = \delta) = \Pr(U = q) = 0$, so the deviation payoff is $\mathbb{E}[(F^{-1}(U) - \delta)\mathbb{1}\{U < q\}] = \Phi_\mu(q) - \delta q = K$. For $\delta > C(1)$ the deviator always wins and earns $m - \delta < K$. $\square$

Step 1: the lower bound. Fix any identical play $\gamma \in \mathcal{BCC}\mathcal{E}(\mu)$. For $\delta \in [0, \bar{v})$ and $\delta' \in (\delta, \bar{v}]$, (4) implies $K \geq \mathbb{E}[(v - \delta')\mathbb{1}\{c < \delta'\}]$. Letting $\delta' \downarrow \delta$ and applying dominated convergence,

$$K \geq \mathbb{E}[(v - \delta)\mathbb{1}\{c \leq \delta\}] \quad \text{for every } \delta \in [0, \bar{v}], \quad (18)$$

where at $\delta = \bar{v}$ the right side is $\mathbb{E}[v - \bar{v}] \leq 0$. Lemma 6 with $X = c$ and $\Delta = [0, \bar{v}]$ then gives

$$R = \mathbb{E}[c] \geq \mathcal{R}_\mu(K). \quad (19)$$

Together with $R = m - NK$ this gives $g_N(K) \geq 0$, where $g_N(K) := m - NK - \mathcal{R}_\mu(K)$.

Lemma 8 (Concavity and uniqueness). $\mathcal{R}_\mu$ is convex, nonincreasing and differentiable on $(0, m)$ with $\mathcal{R}'_\mu(K) = \log q_K$, where $q_K := \Phi_\mu^{-1}(K)$ is the unique solution of $\Phi_\mu(q_K) = K$. Hence g_N is concave on $[0, m]$, with $g_N(0) = m - \int_0^1 \frac{\Phi_\mu(q)}{q} dq > 0$ and $g_N(m) = -(N-1)m < 0$, so g_N has a unique zero $K^* \in (0, m)$, is positive on $[0, K^*)$ and negative on $(K^*, m]$.

Proof. Φ_μ is continuous and strictly increasing, so $\{q : \Phi_\mu(q) > K\} = (q_K, 1]$ and $\mathcal{R}_\mu(K) = \int_{q_K}^1 \frac{\Phi_\mu(q) - K}{q} dq$. The integrand vanishes at q_K , so differentiating under the integral gives $\mathcal{R}'_\mu(K) = -\int_{q_K}^1 \frac{dq}{q} = \log q_K \leq 0$. Since q_K is nondecreasing in K , $\mathcal{R}'_\mu$ is nondecreasing, so $\mathcal{R}_\mu$ is convex and g_N concave. For the endpoints, $\frac{\Phi_\mu(q)}{q} = \mathbb{E}[v \mid v \leq F^{-1}(q)] < m$ for $q < 1$, since μ is non-degenerate under Assumption 1, giving $g_N(0) > 0$, and $\mathcal{R}_\mu(m) = 0$. A concave function positive at the left endpoint and negative at the right endpoint has exactly one zero and crosses from above. $\square$

By Lemma 8, $g_N(K) \geq 0$ forces $K \leq K^*$ and therefore $R = m - NK \geq m - NK^*$, which is the required lower bound.

Step 2: attainment. Consider the identical play distribution with common bid $c = \psi(F(v))$, which is $C(U)$ with $U = F(v)$ uniform, $C = \psi$, and $q_0 = q_{K^*}$. The value marginal is μ , revenue is $\int_0^1 \psi = \mathcal{R}_\mu(K^*) = m - NK^*$, and the rent is K^* . Since $\psi(q_{K^*}^+) = 0$, Lemma 7 covers every deviation $\delta > 0$, and the constraint binds for $\delta \in (0, m - K^*]$. At $\delta = 0$ the deviator ties with the atom at zero and wins with probability $1/N$, earning $\frac{1}{N}\mathbb{E}[v\mathbb{1}\{c = 0\}] = \Phi_\mu(q_{K^*})/N = K^*/N < K^*$. Hence the coupling is a BCCE, and $CRG(\mu) = m - NK^* = \mathcal{R}_\mu(K^*)$, which is strictly positive because $K^* < m$. Note also that $\psi(q) \leq F^{-1}(q)$ pointwise, since $\Phi_\mu(q) \leq qF^{-1}(q)$: the worst-case bid never exceeds the value, as required, since a bidder pooled above the value could profitably commit to zero. $\square$

Proof of Proposition 2. With $g_N(K) = m - NK - \mathcal{R}_\mu(K)$, Lemma 8 gives a unique root crossing from above. Since $g_{N+1} < g_N$ pointwise on $(0, m]$, the root falls, $K_{N+1}^* < K_N^*$. As $\mathcal{R}_\mu$ is strictly decreasing on $[0, m)$ and $CRG = \mathcal{R}_\mu(K_N^*)$, the guarantee rises. $\square$

The strict coarseness gap

Proof of Proposition 4. Let σ_B be a BCE attaining $RG_N(\mu)$ whose two highest bids differ with positive probability. Since $\mathcal{BCE} \subseteq \mathcal{BCCE}$, σ_B is a BCCE with revenue

$RG_N(\mu)$, and Proposition 1 yields an identical play BCCE with revenue strictly below $RG_N(\mu)$. Hence $CRG_N(\mu) < RG_N(\mu)$. $\square$

C Asymptotics

Proof of Theorem 2. From $CRG_N = m - NK_N^* \geq 0$ we get $K_N^* \leq m/N \rightarrow 0$. The integrand in (9) satisfies $0 \leq \frac{(\Phi_\mu(q) - K)^+}{q} \leq \frac{\Phi_\mu(q)}{q} \leq \bar{v}$ and converges pointwise to $\frac{\Phi_\mu(q)}{q}$ as $K \downarrow 0$ whenever $\Phi_\mu(q) > 0$, which holds for every $q > 0$ under Assumption 1. Dominated convergence gives $CRG_N = \mathcal{R}_\mu(K_N^*) \rightarrow \int_0^1 \frac{\Phi_\mu(q)}{q} dq$, and substituting $q = F(v)$ turns this into $\int_0^{\bar{v}} G(v) f(v) dv$, the limiting revenue guarantee of Bergemann, Brooks and Morris (2017). $\square$

Lemma 9 (The interim guarantee converges at rate $1/N$). *For every $N \geq 2$, $0 \leq R_\infty - RG_N \leq \frac{2\bar{v}}{N-1}$.*

Proof. Write $\alpha = \frac{N-1}{N}$. Substituting $t = uF(v)$ in the two conditional means gives

$$G(v) = \int_0^1 F^{-1}(uF(v)) du, \quad \beta_N(v) = \int_0^1 F^{-1}(uF(v)) \alpha u^{\alpha-1} du,$$

so $G(v) - \beta_N(v) = \int_0^1 F^{-1}(uF(v)) (1 - \alpha u^{\alpha-1}) du$. The weight $1 - \alpha u^{\alpha-1}$ integrates to zero, is negative for $u < u_0 := (1 - \frac{1}{N})^N$ and positive above, and $F^{-1}(uF(v))$ is nondecreasing in u , so the integral is nonnegative by the Chebyshev integral inequality: $\beta_N \leq G$ pointwise. For the upper bound, $|G - \beta_N| \leq \bar{v} \int_0^1 |1 - \alpha u^{\alpha-1}| du = 2\bar{v} (u_0^\alpha - u_0) = 2\bar{v} \frac{u_0}{N-1} \leq \frac{2\bar{v}}{N-1}$, using $u_0^{\alpha-1} = (1 - \frac{1}{N})^{-1}$. Integrating against f gives the claim. $\square$

Proof of Proposition 3. The bound $0 \leq R_\infty - RG_N \leq 2\bar{v}/(N-1)$ is Lemma 9. Write $K_N := K_N^*$ for the rent at market size N and $q_N := \Phi_\mu^{-1}(K_N)$ for the pooling quantile, so that $\Phi_\mu(q_N) = K_N$. The proof of Theorem 2 gives $R_\infty = \int_0^1 \frac{\Phi_\mu(q)}{q} dq$ and $K_N \leq m/N \rightarrow 0$, hence $q_N \rightarrow 0$.

Step 1: an exact decomposition. Since $\Phi_\mu(q) \geq K_N$ exactly when $q \geq q_N$, (9) reads $CRG_N = \int_{q_N}^1 \frac{\Phi_\mu(q) - K_N}{q} dq$, and subtracting from R_∞ ,

$$R_\infty - CRG_N = T_N + K_N \log \frac{1}{q_N}, \quad T_N := \int_0^{q_N} \frac{\Phi_\mu(q)}{q} dq.$$

The first term is the value on the pooled quantiles, and the second is the rent on the bidding quantiles.

Step 2: the order of the rent. By (10), $NK_N = m - CRG_N \rightarrow m - R_\infty$, and this limit is strictly positive because $\Phi_\mu(q)/q$, the average of the strictly increasing function F^{-1} over $[0, q]$, is strictly below $F^{-1}(q)$, so $R_\infty < \int_0^1 F^{-1} = m$. Hence $K_N = \frac{m-R_\infty}{N}(1+o(1))$ and $\log \frac{1}{K_N} = \log N + O(1)$.

Step 3: the order of the pooling quantile. This is the only step that uses the continuity and positivity of f at 0 in Assumption 1. They give $F(x) = f(0)x(1+o(1))$ as $x \downarrow 0$, hence $F^{-1}(t) = \frac{t}{f(0)}(1+o(1))$ as $t \downarrow 0$ and, integrating, $\Phi_\mu(q) = \frac{q^2}{2f(0)}(1+o(1))$ as $q \downarrow 0$. At $q = q_N$ this reads $K_N = \frac{q_N^2}{2f(0)}(1+o(1))$, so

$$\log \frac{1}{q_N} = \frac{1}{2} \log \frac{1}{K_N} + O(1) = \frac{1}{2} \log N + O(1).$$

The square root in $q_N \asymp \sqrt{K_N}$ is the source of the factor $\frac{1}{2}$ in the proposition.

Step 4: the pooled term is of smaller order. Since $\Phi_\mu(q)/q \leq F^{-1}(q)$, $0 \leq T_N \leq \int_0^{q_N} F^{-1}(q) dq = \Phi_\mu(q_N) = K_N = O(1/N)$.

Step 5: conclusion. By Steps 2 and 3, $K_N \log \frac{1}{q_N} = \frac{m-R_\infty}{N}(1+o(1))(\frac{1}{2} \log N + O(1)) = \frac{(m-R_\infty) \log N}{2N}(1+o(1))$, and adding $T_N = O(1/N) = o(\log N/N)$ gives the expansion for $R_\infty - CRG_N$. Finally $RG_N - CRG_N = (R_\infty - CRG_N) - (R_\infty - RG_N)$, and the second bracket is $O(1/N) = o(\log N/N)$ by Lemma 9, so $RG_N - CRG_N$ has the same expansion. $\square$

D Reserve Prices

Proof of Proposition 5

The proof follows Lemmas 3 and 4, and I describe only what changes. Bidding in $[0, r)$ is the same as bidding 0, and I use this normalization throughout. Lemma 3 applies unchanged, so take $\hat{\sigma}$ exchangeable. The pooling kernel of Lemma 4 is replaced by the kernel Define the Markov kernel Γ_r on $V \times A$ by

$$\Gamma_r((v, a), \cdot) = \begin{cases} \delta_{\{v\}} \otimes \delta_{\{\mathbf{0}\}}, & a_{(1)} < r, \\ \delta_{\{v\}} \otimes \left[\frac{N-1}{N} \delta_{\{\mathbf{a}_{(1)}\}} + \frac{1}{N} \delta_{\{\max\{\mathbf{a}_{(2)}, r\}\}} \right], & a_{(1)} \geq r \text{ and } (a_{(2)} \geq r \text{ or } v \geq a_{(1)}), \\ \delta_{\{v\}} \otimes \left[\frac{N-1}{N} \delta_{\{\mathbf{a}_{(1)}\}} + \frac{1}{N} \delta_{\{\mathbf{0}\}} \right], & a_{(1)} \geq r > a_{(2)} \text{ and } v < a_{(1)}, \end{cases}$$

and let γ be the image of $\widehat{\sigma}\Gamma_r$ under one further normalization: on the event where the common bid equals r and $v < r$, replace the common bid by $\mathbf{0}$. The γ is still an exchangeable identical play distribution with value marginal μ . By construction, it is almost surely that $v \geq r$ when the common bid equals r . Write c for the common bid under γ . The proposition is then proved by the following three claims.

Revenue and payoffs. On the first branch nothing is sold before or after. On the second branch the common bid is a convex combination of $a_{(1)}$ and $\max\{a_{(2)}, r\} \leq a_{(1)}$, so revenue weakly falls and, by (13), the buyers' total payoff weakly rises, exactly as in Lemma 4. On the third branch, and on the states changed by the final normalization, a sale at a price above the value is replaced by withholding, with probability $\frac{1}{N}$ on the third branch and with probability one under the normalization, which lowers revenue and raises the total payoff from a negative number toward zero. By exchangeability each buyer's payoff is $1/N$ of the total, so $K_\gamma \geq K_{\widehat{\sigma}}$.

Deviations. A deviation to $\delta < r$ never wins under either distribution and earns zero, while $K_\gamma \geq 0$ because deviating to 0 is available. Fix $\delta > r$ that is not an atom of the law of c . An atom of the averaged law of M_i at δ requires an atom of $a_{(1)}$ or $a_{(2)}$ there, which the kernel maps to an atom of c at the same point, so ties have probability zero under both distributions, and it suffices to compare $\mathbb{E}_\gamma[(v - \delta)\mathbb{1}\{c < \delta\}]$ with $\mathbb{E}_{\widehat{\sigma}}[(v - \delta)\mathbb{1}\{M_i < \delta\}]$ averaged over i . Couple the two through the randomization in the kernel, averaging over buyers as in Step 1 of Lemma 4. On the second branch with $a_{(2)} \geq r$ the kernel is that of Lemma 4, and Step 1 of that lemma gives the equality of the conditional laws of M_i and c . On every other branch, and after the final normalization, whenever the coupled pair (M_i, c) differs both coordinates are at most $r < \delta$, so both indicators equal one. Hence $\mathbb{E}_\gamma[(v - \delta)\mathbb{1}\{c < \delta\}] = \mathbb{E}_{\widehat{\sigma}}[(v - \delta)\mathbb{1}\{M_i < \delta\}] \leq K_{\widehat{\sigma}} \leq K_\gamma$ for every such δ .

For an atom of c in $(r, \bar{v}]$, both one-sided limits are available within D and the argument of Lemma 2 applies without change. The deviation to exactly r is the one level not covered by that argument, since there are no admissible deviations below the reserve from which to take the upward limit, and it is checked directly. A deviator bidding r wins when $c = 0$ and ties the full pool when $c = r$, so the deviation payoff is $(1 - \frac{1}{N})A + \frac{1}{N}B$ with $A = \mathbb{E}_\gamma[(v - r)\mathbb{1}\{c = 0\}]$ and $B = \mathbb{E}_\gamma[(v - r)\mathbb{1}\{c \leq r\}]$. Letting $\delta \downarrow r$ through D in the display above gives $K_\gamma \geq B$ by dominated convergence, and the normalization placed $v \geq r$ on $\{c = r\}$, so $B - A = \mathbb{E}_\gamma[(v - r)\mathbb{1}\{c = r\}] \geq 0$ and the deviation payoff is at most $B \leq K_\gamma$. Hence γ is a BCCE. $\square$

The two-pool family is feasible

Lemma 10 (Feasibility). *Fix r , and let (s, q_0) satisfy (14)–(15) with $K = \Phi_\mu(q_0) - rq_0 \geq 0$ and $\Phi_\mu(q_0) - \Phi_\mu(s) \geq r(q_0 - s)$. Then the comonotone identical play distribution with common bid $c = 0$ on the lowest s quantiles of v , $c = r$ on quantiles $[s, q_0)$, and $c = \psi_r(q) := (\Phi_\mu(q) - K)/q$ at quantile $q \geq q_0$ is a BCCE of the auction with reserve r , with revenue R as in (15).*

Proof. The construction has value marginal μ , and by (13) and (15) each buyer's payoff is exactly K . The common bid is $C(U)$ with $U = F(v)$, where $C = 0$ on $[0, s)$, $C = r$ on $[s, q_0)$, and $C = \psi_r$ on $[q_0, 1]$, with $\psi_r(q_0) = r$ by (14). Lemma 7 covers every deviation $\delta > r$. For $\delta \in (0, r)$ the deviator never clears the reserve and earns $0 \leq K$. For $\delta = 0$ the deviator ties the withholding pool below the reserve, no sale occurs, and the payoff is again $0 \leq K$. For $\delta = r$ the deviator wins the withholding pool and ties the reserve pool, earning

$$\begin{aligned} (\Phi_\mu(s) - rs) + \frac{1}{N}(\Phi_\mu(q_0) - \Phi_\mu(s) - r(q_0 - s)) \\ = K - (1 - \frac{1}{N})(\Phi_\mu(q_0) - \Phi_\mu(s) - r(q_0 - s)) \leq K, \end{aligned}$$

where the identity substitutes (14) and the inequality is the feasibility condition. Hence the construction is a BCCE, and $CRG_N(r) \leq \widehat{V}_N(r)$. $\square$

Comonotonicity is without loss

Lemma 11 (Comonotonization). *For every identical play BCCE of the auction with reserve r there is an identical play BCCE with the same bid marginal, hence the same revenue, in which the common bid is $c = C(F(v))$ for C the quantile function of that marginal.*

Proof. Let (v, c) be the given joint law with bid marginal F_c , and let (v, c°) with $c^\circ = C(F(v))$ be the comonotone coupling of the same marginals. Revenue depends on the joint law only through F_c and is unchanged. Write $s = \Pr(c = 0)$ and $W = \mathbb{E}[v\mathbb{1}\{c = 0\}]$. Under the comonotone coupling the event $\{c^\circ = 0\}$ coincides up to null sets with the lower set $\{F(v) < s\}$, so $W^\circ = \Phi_\mu(s) \leq W$ by Lemma 5, and the accounting identity (13) gives $K^\circ = (m - W^\circ - R)/N \geq K$ for the payoff.

Every deviation payoff weakly falls. For $\delta \in [0, r)$ it is zero under both couplings. For $\delta \geq r$, the payoff to the fixed bid δ is $(1 - \frac{1}{N})A_\delta + \frac{1}{N}B_\delta$ with $A_\delta = \mathbb{E}[(v - \delta)\mathbb{1}\{c < \delta\}]$

and $B_\delta = \mathbb{E}[(v - \delta)\mathbb{1}\{c \leq \delta\}]$. The probabilities $\Pr(c < \delta)$ and $\Pr(c \leq \delta)$ are fixed by the marginal, and Lemma 5 says the comonotone coupling minimizes both $\mathbb{E}[v\mathbb{1}\{c < \delta\}]$ and $\mathbb{E}[v\mathbb{1}\{c \leq \delta\}]$ among couplings with those probabilities, so $A_\delta^\circ \leq A_\delta$ and $B_\delta^\circ \leq B_\delta$. Every coarse constraint therefore holds at the comonotone coupling, with a weakly larger left side and a weakly smaller right side. $\square$

The chain as an envelope

Lemma 12 (Envelope form). *Let γ be a comonotone identical play BCCE with payoff K , withholding share s , and bid quantile function C . Then*

$$C(q) \geq \underline{C}_K(q) := \max \left\{ r, \frac{\Phi_\mu(q) - K}{q} \right\} \quad \text{for every } q \in (s, 1].$$

Conversely, any comonotone identical play distribution with payoff $K \geq 0$ whose quantile function dominates $\underline{C}_K$ on $(s, 1]$ satisfies $K \geq \mathbb{E}[(v - \delta)\mathbb{1}\{c \leq \delta\}]$ for every $\delta \geq r$.

Proof. Necessity. On $(s, 1]$ the codomain gives $C \geq r$. The closed form of the constraint, obtained as in (18), holds for every $\delta \geq r$, so Lemma 6 with $X = c$ and $\Delta = [r, \bar{v}]$ gives $C(q) \geq (\Phi_\mu(q) - K)/q$ whenever $C(q) \geq r$, in particular on $(s, 1]$. Sufficiency. Fix $\delta \geq r$ and let $x = F_c(\delta)$. Every $q < x$ has $C(q) \leq \delta$, hence $(\Phi_\mu(q) - K)/q \leq \underline{C}_K(q) \leq C(q) \leq \delta$. Letting $q \uparrow x$ and using continuity of Φ_μ gives $\Phi_\mu(x) - \delta x \leq K$, which is the stated inequality. $\square$

Proof of Theorem 3: exactness

For $r \geq m$, the distribution in which every buyer always bids 0 is a BCCE, since a deviation to any $\delta \geq r$ wins and earns $m - \delta \leq 0$, so the guarantee is zero. For $r = 0$ the family constraint (14) reads $K = \Phi_\mu(q_0)$ and, at $s = 0$, (15) is the system (10), so the claim is Theorem 1 itself. Fix now $r \in (0, m)$. For $s \in [0, 1)$ and $\kappa \in [0, m]$ define

$$h_s(\kappa) := m - \Phi_\mu(s) - N\kappa - \int_s^1 \underline{C}_\kappa(q) dq.$$

Step 1: h_s is concave. The function $q \mapsto \Phi_\mu(q) - rq$ vanishes at 0, decreases on $[0, F(r)]$ and increases thereafter, so its sublevel set at any $\kappa \geq 0$ is an interval

$[0, q_r(\kappa)]$ with q_r nondecreasing, and $(\Phi_\mu(q) - \kappa)/q > r$ exactly for $q > q_r(\kappa)$. Hence $h'_s(\kappa) = -N + \log(1/(s \vee q_r(\kappa)))$, which is nonincreasing in κ .

Step 2: the share bound. Let γ be any comonotone identical play BCCE with share s , payoff K and revenue R . Comonotonicity gives $W = \Phi_\mu(s)$, so (13) reads $R = m - \Phi_\mu(s) - NK$, while Lemma 12 gives $R = \int_s^1 C \geq \int_s^1 \underline{C}_K$. Subtracting, $h_s(K) \geq 0$. By Step 1 the set $\{\kappa \in [0, m] : h_s(\kappa) \geq 0\}$ is an interval, nonempty since it contains K , and since $h_s(m) \leq m - Nm < 0$ its right endpoint $\kappa_+(s)$ is a largest root. So $K \leq \kappa_+(s)$ and

$$R \geq \mathcal{F}(s) := m - \Phi_\mu(s) - N\kappa_+(s).$$

Step 3: admissible shares. The equilibrium's share also satisfies $\Phi_\mu(s) - rs \leq K$. Otherwise, by continuity there is $\eta > 0$ with $(\Phi_\mu(q) - K)/q > r$ on $(s, s + \eta)$, so the envelope forces $C > r$ there, there is no bid atom at r , and $F_c(r) = s$. The chain at $\delta = r$ then reads $K \geq \Phi_\mu(s) - rs$, a contradiction. Call s *admissible* if h_s has a root in $[0, m]$ and $\Phi_\mu(s) - rs \leq \kappa_+(s)$. Every equilibrium share is admissible, so Proposition 5 and Lemma 11 give

$$CRG_N(r) \geq \inf \{ \mathcal{F}(s) : s \text{ admissible} \}.$$

Step 4: candidates and the family. First note that $\kappa_+(s) < m - r$ for every s . At $\kappa = m - r$ the envelope equals r throughout $(s, 1]$, since $(\Phi_\mu(q) - \kappa)/q \leq m - \kappa = r$, so $h_s(m - r) = rs - \Phi_\mu(s) - (N - 1)(m - r)$. The map $s \mapsto rs - \Phi_\mu(s)$ is maximized where $F^{-1}(s) = r$, so $h_s(m - r) \leq rF(r) - \Phi_\mu(F(r)) - (N - 1)(m - r) < 0$ by the hypothesis of the theorem, and since $h'_s(\kappa) = -N$ for $\kappa > m - r$, no root lies at or beyond $m - r$. Hence the crossing $q_r(\kappa_+(s))$ is interior. For admissible s , consider the comonotone distribution with bids 0 on the lowest s quantiles and $\underline{C}_{\kappa_+(s)}$ above. Admissibility says $\kappa_+(s) \geq \Phi_\mu(s) - rs$, equivalently $q_0 := q_r(\kappa_+(s)) \geq s$, so the bids equal r on (s, q_0) and $(\Phi_\mu(q) - \kappa_+(s))/q$ above. By the definition of q_r and continuity, $\Phi_\mu(q_0) - rq_0 = \kappa_+(s)$, which is (14), while $h_s(\kappa_+(s)) = 0$ is (15). The candidate is therefore a member of the two-pool family with revenue $\mathcal{F}(s)$. Conversely, every family member at parameters (s, q_0) has payoff $K = \Phi_\mu(q_0) - rq_0$, which its feasibility condition makes at least $\Phi_\mu(s) - rs$, and K is a root of h_s by (15), so s is admissible and the member's revenue is at least $\mathcal{F}(s)$. Hence $\widehat{V}_N(r) = \inf \{ \mathcal{F}(s) : s \text{ admissible} \}$.

Step 5: attainment and the tie constraint. Lemma 10 shows the candidate at s

is a BCCE whenever the average value in the reserve pool is at least r , which holds whenever $F^{-1}(s) \geq r$, that is $s \geq F(r)$. Shares below $F(r)$ are dominated. For $s \leq s' \leq F(r)$,

$$h_{s'}(\kappa) - h_s(\kappa) = \int_s^{s'} (\underline{C}_\kappa(q) - F^{-1}(q)) dq \geq \int_s^{s'} (r - F^{-1}(q)) dq \geq 0,$$

so $\kappa_+(s') \geq \kappa_+(s)$ and $\mathcal{F}(s) - \mathcal{F}(s') = (\Phi_\mu(s') - \Phi_\mu(s)) + N(\kappa_+(s') - \kappa_+(s)) \geq 0$. Moreover, admissibility passes from s to s' , since $\Phi_\mu(s') - rs' \leq \Phi_\mu(s) - rs \leq \kappa_+(s) \leq \kappa_+(s')$. The infimum over admissible shares is therefore unchanged when restricted to $s \geq F(r)$, and there every candidate is a BCCE, so $CRG_N(r) \leq \inf\{\mathcal{F}(s) : s \text{ admissible}\}$. Combining with Steps 3 and 4, $CRG_N(r) = \hat{V}_N(r)$. $\square$

Proof of Theorem 3: continuity at a zero reserve

Write $h(s, \kappa; r)$ for the function $h_s(\kappa)$ above with the reserve made explicit, so that $\underline{C}_\kappa(q) = \max\{r, (\Phi_\mu(q) - \kappa)/q\}$ and

$$h(s, \kappa; 0) = g_N(\kappa) - \Phi_\mu(s) + \int_0^s \frac{(\Phi_\mu(q) - \kappa)^+}{q} dq,$$

and write $\kappa_+(s; r)$, $\mathcal{F}(s; r)$, and admissibility at r accordingly. Three facts are used. First, h is continuous in (s, κ, r) and, since $\max\{0, x\} \leq \max\{r, x\} \leq \max\{0, x\} + r$, satisfies $h(s, \kappa; 0) - r \leq h(s, \kappa; r) \leq h(s, \kappa; 0)$. Hence $\kappa_+(s; r) \leq \kappa_+(s; 0)$ and $\mathcal{F}(s; r) \geq \mathcal{F}(s; 0)$ for every s admissible at r . Also $h(s, \kappa; 0) \leq g_N(\kappa)$, since $\int_0^s (\Phi_\mu(q) - \kappa)^+/q dq \leq \int_0^s F^{-1} = \Phi_\mu(s)$, so $\kappa_+(s; 0) \leq K^*$. Second, $\kappa_+(\cdot; 0)$ is upper semi-continuous: if $s_n \rightarrow s$ and $\kappa_+(s_n; 0) \rightarrow \kappa$ along a subsequence, then $h(s, \kappa; 0) = \lim_n h(s_n, \kappa_+(s_n; 0); 0) \geq 0$, so $\kappa \leq \kappa_+(s; 0)$. Third, every s admissible at $r = 0$ has $\mathcal{F}(s; 0) \geq \mathcal{F}(0; 0) = CRG_N(0)$. Indeed, let $\kappa = \kappa_+(s; 0)$. Admissibility at $r = 0$ reads $\Phi_\mu(s) \leq \kappa$, so $s \leq q_\kappa$, the integral in $h(s, \kappa; 0)$ vanishes, and $h(s, \kappa; 0) = 0$ gives $g_N(\kappa) = \Phi_\mu(s) \geq 0$. Since $-g'_N$ is nondecreasing with $-g'_N(K^*) = N - \log(1/q_{K^*}) \leq N$, $g_N(\kappa) = g_N(\kappa) - g_N(K^*) \leq N(K^* - \kappa)$. Therefore $\mathcal{F}(s; 0) = m - \Phi_\mu(s) - N\kappa = m - g_N(\kappa) - N\kappa \geq m - NK^*$.

Upper semicontinuity. The share $s = 0$ is admissible at every $r < g_N(0)$, since then $h(0, 0; r) \geq g_N(0) - r > 0$, and $\mathcal{F}(0; r) = m - N\kappa_+(0; r)$. By the first fact, $|h(0, \kappa; r) - g_N(\kappa)| \leq r$ uniformly in κ . The slope $g'_N(K^*)$ is strictly negative, since a zero slope at

a root of a concave function would force $g_N \leq 0$ everywhere, contradicting $g_N(0) > 0$, so g_N is positive on $[0, K^*)$ and decreasing on $[K^*, m]$. Hence for every $\eta > 0$ and all small r , $h(0, \cdot; r) > 0$ at $K^* - \eta$ and $h(0, \cdot; r) < 0$ on $[K^* + \eta, m]$, so $\kappa_+(0; r) \rightarrow K^*$, $\mathcal{F}(0; r) \rightarrow CRG_N(0)$, and $\limsup_{r \downarrow 0} \widehat{V}_N(r) \leq CRG_N(0)$.

Lower semicontinuity. Take $r_n \downarrow 0$ and, for each n , a share s_n admissible at r_n with $\mathcal{F}(s_n; r_n) \leq \widehat{V}_N(r_n) + r_n$. Admissibility and the first fact give $\Phi_\mu(s_n) - r_n s_n \leq \kappa_+(s_n; r_n) \leq \kappa_+(s_n; 0) \leq K^*$, so the s_n stay in a compact subset of $[0, 1]$. Passing to a subsequence, $s_n \rightarrow \bar{s}$ and $\kappa_+(s_n; 0) \rightarrow \bar{\kappa}$, and in the limit $\Phi_\mu(\bar{s}) \leq \bar{\kappa} \leq \kappa_+(\bar{s}; 0)$ by the second fact, so $\bar{s}$ is admissible at $r = 0$. By the first and third facts,

$$\liminf_n \widehat{V}_N(r_n) \geq \liminf_n \mathcal{F}(s_n; r_n) \geq \liminf_n \mathcal{F}(s_n; 0) = m - \Phi_\mu(\bar{s}) - N\bar{\kappa} \geq \mathcal{F}(\bar{s}; 0) \geq CRG_N(0).$$

Since $CRG_N(r) = \widehat{V}_N(r)$ for $r > 0$ by the exactness part, $\lim_{r \downarrow 0} CRG_N(r) = CRG_N(0)$. $\square$

E Symmetric Priors

Throughout, π is exchangeable on V^N with per-buyer marginal μ , a BCCE is a joint law σ of $(v_1, \dots, v_N, a_1, \dots, a_N)$ with value marginal π satisfying the coarse constraints, $u_i = q_i(a)(v_i - a_i)$, $M_i := \max_{j \neq i} a_j$, and $K_i := \mathbb{E}_\sigma[u_i]$. Since the good is always allocated, $\sum_i u_i = v_{\text{win}} - a_{(1)}$, where v_{win} is the winner's value.

Symmetrization

Lemma 13 (Symmetrization under exchangeable priors). *For a BCCE σ let $\widehat{\sigma} := \frac{1}{N!} \sum_\rho \sigma^\rho$, where σ^ρ is the image of σ under $(v, a) \mapsto (v_\rho, a_\rho)$ with $(v_\rho)_i = v_{\rho(i)}$ and $(a_\rho)_i = a_{\rho(i)}$. Then $\widehat{\sigma}$ is an exchangeable BCCE with $R(\widehat{\sigma}) = R(\sigma)$ and $K_i = K$ for every i .*

Proof. The proof of Lemma 3 applies with values permuted together with bids. The value marginal of σ^ρ is π by exchangeability of π , buyer i 's equilibrium and deviation payoffs under σ^ρ are buyer $\rho(i)$'s under σ since $u_i(v_\rho, a_\rho) = u_{\rho(i)}(v, a)$ and $M_i(a_\rho) = M_{\rho(i)}(a)$, and $a_{(1)}$ is permutation invariant. Averaging over ρ gives $K_i = K$. $\square$

Proof of Theorem 4

The fixed point. Let $\tilde{g}(K) := m_N - NK - \mathcal{R}_\mu(K) = g_N(K) + (m_N - m)$. By Lemma 8, $\tilde{g}$ is concave on $[0, m]$ with $\tilde{g}(0) \geq g_N(0) > 0$, and $\tilde{g}(m) = m_N - Nm \leq 0$ since $\mathcal{R}_\mu(m) = 0$ and $\max_i v_i \leq \sum_i v_i$. Hence $\tilde{g}$ has a unique zero $\tilde{K} \in (0, m]$, is positive on $[0, \tilde{K})$ and negative on $(\tilde{K}, m]$, and $\tilde{K} \geq K^*$ with strict inequality if and only if $m_N > m$. The following properties of $\tilde{q} = \Phi_\mu^{-1}(\tilde{K})$ will be needed. First, $\mathcal{R}_\mu(\tilde{K}) = \int_{\tilde{q}}^1 \frac{\Phi_\mu(q) - \tilde{K}}{q} dq$, since $\Phi_\mu \geq \tilde{K}$ exactly on $[\tilde{q}, 1]$. Second, because $\Phi_\mu(q)/q$ is the average of F^{-1} on $[0, q]$ and hence at most $F^{-1}(q)$,

$$m_N - N\tilde{K} = \mathcal{R}_\mu(\tilde{K}) \leq \int_{\tilde{q}}^1 F^{-1}(q) dq - \tilde{K} \log \frac{1}{\tilde{q}} = m - \tilde{K} - \tilde{K} \log \frac{1}{\tilde{q}},$$

so that

$$\tilde{K} (N - 1 + \log \tilde{q}) \geq m_N - m \geq 0, \quad (20)$$

with strict inequality on the right when $m_N > m$. In particular $N + \log \tilde{q} \geq 1$.

The lower bound. Fix a BCCE σ with revenue $R = \mathbb{E}[a_{(1)}]$. For each buyer i , the coarse constraint at a fixed bid δ' that is not an atom of the law of M_i reads $K_i \geq \mathbb{E}[(v_i - \delta') \mathbb{1}\{M_i < \delta'\}]$, and letting $\delta' \downarrow \delta$ through non-atoms gives $K_i \geq \mathbb{E}[(v_i - \delta) \mathbb{1}\{M_i \leq \delta\}]$ for every $\delta \in [0, \bar{v}]$. Lemma 6 applied to the pair (v_i, M_i) , which uses only the marginal $v_i \sim \mu$, gives

$$\mathbb{E}[M_i] \geq \mathcal{R}_\mu(K_i) \quad \text{for every buyer } i. \quad (21)$$

Since $M_i \leq a_{(1)}$ pointwise, $R \geq \mathbb{E}[M_i]$. Since the winner's value is at most the highest value, $\sum_i K_i = \mathbb{E}[v_{\text{win}} - a_{(1)}] \leq m_N - R$. Let $K_{\min} := \min_i K_i \leq \frac{1}{N} \sum_i K_i$. Then $m_N - NK_{\min} \geq m_N - \sum_i K_i \geq R \geq \mathcal{R}_\mu(K_{\min})$, that is, $\tilde{g}(K_{\min}) \geq 0$, so $K_{\min} \leq \tilde{K}$, and since $\mathcal{R}_\mu$ is nonincreasing, $R \geq \mathcal{R}_\mu(K_{\min}) \geq \mathcal{R}_\mu(\tilde{K}) = m_N - N\tilde{K}$. Taking the infimum over BCCE gives the inequality.

Equality. If $m_N = m$ then $\max_i v_i = v_i$ almost surely for every i , so the prior is common-valued and the claim is Theorem 1. Assume $m_N > m$, so that $N + \log \tilde{q} > 1$ by (20). Write $P := \{\max_i v_i \leq \tilde{v}\}$ for the pooling event. The agreement hypothesis implies $P = \{v_i \leq \tilde{v}\}$ for every i up to null sets, since $v_i > \tilde{v}$ forces $v_j = v_i > \tilde{v}$ for all j , and on P^c the values are common. Write v for the common value there. The non-degeneracy hypothesis says that with positive probability two buyers have

positive values on P , which is equivalent to

$$A := \frac{1}{N} \mathbb{E}[\max_j v_j \mathbb{1}_P] < \mathbb{E}[v_i \mathbb{1}_P] = \Phi_\mu(\tilde{q}) = \tilde{K}, \quad (22)$$

where the middle equality uses $P = \{v_i \leq \tilde{v}\}$ and the first uses $\sum_j v_j \geq \max_j v_j$ on P with strict inequality on a set of positive probability.

For $\varepsilon > 0$ define

$$U_\varepsilon := \tilde{K} - \frac{\varepsilon \tilde{q}}{N + \log \tilde{q}}, \quad \tilde{c}_\varepsilon(q) := \frac{\Phi_\mu(q) - U_\varepsilon}{q} \quad \text{for } q \in (\tilde{q}, 1],$$

and consider the following exchangeable distribution σ_ε with value marginal π . On P , a buyer whose value is maximal, chosen uniformly at random among the maximizers, bids ε and every other buyer bids 0. Call this buyer *designated*. On P^c every buyer bids $\tilde{c}_\varepsilon(F(v))$. Since $U_\varepsilon < \tilde{K}$, $\tilde{c}_\varepsilon$ is positive on $(\tilde{q}, 1]$ and strictly increasing there, with $\tilde{c}_\varepsilon(\tilde{q}) = (\tilde{K} - U_\varepsilon)/\tilde{q} = \varepsilon/(N + \log \tilde{q}) =: \eta_\varepsilon \in (0, \varepsilon)$ and $\tilde{c}_\varepsilon(1) = m - U_\varepsilon$.

Revenue and rent. Revenue is $R_\varepsilon = \varepsilon \tilde{q} + \int_{\tilde{q}}^1 \tilde{c}_\varepsilon(q) dq = \varepsilon \tilde{q} + \mathcal{R}_\mu(\tilde{K}) + (\tilde{K} - U_\varepsilon) \log \frac{1}{\tilde{q}}$. On P the good goes to a maximal-value buyer at price ε and on P^c it is allocated by a tie at price $\tilde{c}_\varepsilon$, so the buyers' total payoff is $\mathbb{E}[\max_j v_j \mathbb{1}_P] - \varepsilon \tilde{q} + \mathbb{E}[(v - \tilde{c}_\varepsilon(F(v))) \mathbb{1}_{P^c}] = m_N - R_\varepsilon$, and by exchangeability each buyer's rent is

$$\frac{m_N - R_\varepsilon}{N} = \frac{m_N - \mathcal{R}_\mu(\tilde{K}) - \varepsilon \tilde{q} + (\tilde{K} - U_\varepsilon) \log \tilde{q}}{N} = \tilde{K} - \frac{\varepsilon \tilde{q}}{N} \cdot \frac{N + \log \tilde{q} - \log \tilde{q}}{N + \log \tilde{q}} = U_\varepsilon,$$

using $m_N - \mathcal{R}_\mu(\tilde{K}) = N\tilde{K}$. Hence $R_\varepsilon = m_N - NU_\varepsilon = m_N - N\tilde{K} + \frac{N\varepsilon\tilde{q}}{N + \log \tilde{q}} \rightarrow m_N - N\tilde{K}$ as $\varepsilon \downarrow 0$.

Obedience. Fix a buyer i and write $D(\delta)$ for the payoff from the fixed bid δ . Under σ_ε , buyer i 's highest competing bid is 0 on P when i is designated, ε on P when i is not, and $\tilde{c}_\varepsilon(F(v)) \geq \eta_\varepsilon$ on P^c . By exchangeability $\mathbb{E}[v_i \mathbb{1}_P \mathbb{1}\{i \text{ designated}\}] = A$ and $\mathbb{E}[v_i \mathbb{1}_P \mathbb{1}\{i \text{ not designated}\}] = \tilde{K} - A > 0$. Choose ε small enough that $U_\varepsilon \geq A$ and $\tilde{K} - A \geq 2\varepsilon\tilde{q}$.

For $\delta > \varepsilon$ the check is that of Lemma 7. Buyer i wins on all of P and on the states of P^c with $\tilde{c}_\varepsilon(F(v)) < \delta$, so for $\delta = \tilde{c}_\varepsilon(q) \in (\varepsilon, m - U_\varepsilon]$ the winning event is $\{v_i < F^{-1}(q)\}$ up to null sets and $D(\delta) = \Phi_\mu(q) - \delta q = U_\varepsilon$, while for $\delta > m - U_\varepsilon$ buyer i always wins and $D(\delta) = m - \delta < U_\varepsilon$.

The deviations $\delta \leq \varepsilon$ are where the construction differs from identical play. At $\delta =$

0, buyer i ties only when designated, sharing with $N - 1$ others, so $D(0) \leq A/N \leq U_\varepsilon$. For $\delta \in (0, \eta_\varepsilon]$, buyer i wins exactly the designated states of P , so $D(\delta) \leq A \leq U_\varepsilon$. For $\delta \in (\eta_\varepsilon, \varepsilon]$, write $\delta = \tilde{c}_\varepsilon(q)$ with $q \in (\tilde{q}, 1)$. Buyer i wins the designated states of P and the states of P^c with $v < F^{-1}(q)$, and at $\delta = \varepsilon$ also ties with the designated buyer in the non-designated states of P , winning there with probability at most $\frac{1}{2}$. Since $P \cup \{\tilde{v} < v < F^{-1}(q)\} = \{v_i < F^{-1}(q)\}$ up to null sets and $\Phi_\mu(q) - \delta q = U_\varepsilon$,

$$D(\delta) \leq U_\varepsilon - \frac{1}{2} \mathbb{E}[(v_i - \delta) \mathbb{1}_P \mathbb{1}\{i \text{ not designated}\}] \leq U_\varepsilon - \frac{1}{2}(\tilde{K} - A - \varepsilon \tilde{q}) < U_\varepsilon.$$

So σ_ε is a BCCE for every small ε , and $CRG(\pi) \leq \inf_\varepsilon R_\varepsilon = m_N - N\tilde{K}$. With the lower bound, this proves the equality claim. $\square$

Remark 2 (Non-attainment). For $N = 2$ the infimum in the equality case of Theorem 4 is not attained when the values differ with positive probability on P . Attainment would force equality in $R \geq \mathbb{E}[M_i]$, hence $a_1 = a_2$ almost surely, and equality in $\sum_i K_i \leq m_N - R$, hence a winner with the higher value almost surely. But with tied bids the winner is uniform, so the two values would have to coincide almost surely.

F Atomic Priors, Convex Order, and the Binary Bound

Proof of Theorem 5

For an arbitrary prior, the definitions (8), (9), and (11) are read with the quantile function $F^{-1}(t) = \inf\{x : F(x) \geq t\}$: $\Phi_\mu(q) = \int_0^q F^{-1}(t) dt$ remains convex, nondecreasing, and absolutely continuous, with $\Phi_\mu(0) = 0$ and $\Phi_\mu(1) = m$, and $q_K := \sup\{q : \Phi_\mu(q) \leq K\}$ replaces $\Phi_\mu^{-1}(K)$, with $\Phi_\mu(q_K) = K$ by continuity whenever $q_K < 1$. Proposition 1, Lemma 1, Lemma 2, Lemma 6, and Lemma 7 nowhere use the density, so it suffices to check the three steps of Appendix B.

The rearrangement inequality. Lemma 5 holds for arbitrary μ with the equality clause weakened: $\mathbb{E}[v \mathbb{1}_B] \geq \Phi_\mu(q)$ for every event with $\Pr(B) = q$, with equality if and only if B agrees almost surely with a lower-contour set $\{v \leq F^{-1}(q)\}$, up to splitting an atom at $F^{-1}(q)$. This is the Hardy–Littlewood inequality in its general form. Lemma 6 uses only this inequality and the monotonicity of $q \mapsto (\Phi_\mu(q) - K)/q$, whose derivative computation requires only that F^{-1} be nondecreasing. The lower

bound $R \geq \mathcal{R}_\mu(K)$ therefore holds without change.

Uniqueness. The proof of Lemma 8 never uses the density: the same differentiation gives $\mathcal{R}'_\mu(K) = \log q_K$ with the sup-definition of q_K , so g_N is concave, positive at 0 by non-degeneracy, negative at m , and its zero $K^* \in (0, m)$ is unique.

Attainment. A deterministic bid $c = \psi(F(v))$ no longer suffices, since at an atom of μ the worst case must randomize the bid. Set instead $v = F^{-1}(U)$ and $c = \psi(U)$ with U uniform on $[0, 1]$. The value marginal is μ , revenue is $\int_0^1 \psi = \mathcal{R}_\mu(K^*) = m - NK^*$, and the rent is K^* . Lemma 7, which is stated for a uniform U and applies since $K^* > 0$, covers every deviation $\delta > 0$, and the deviation to zero ties the atom of mass q_{K^*} and earns $\Phi_\mu(q_{K^*})/N = K^*/N$. Hence the coupling is a BCCE and $CRG(\mu) = m - NK^* = \mathcal{R}_\mu(K^*) > 0$. $\square$

Proofs for Section 9

Proof of Proposition 8. Suppose $\mu \preceq_{\text{cx}} \nu$. By the quantile characterization of the convex order (Shaked and Shanthikumar, 2007, Theorem 3.A.5), the two distributions share the mean m and satisfy $\Phi_\nu(q) \leq \Phi_\mu(q)$ for all q . Hence $\mathcal{R}_\nu \leq \mathcal{R}_\mu$ pointwise by (9), so

$$g_N^\nu(K) = m - NK - \mathcal{R}_\nu(K) \geq m - NK - \mathcal{R}_\mu(K) = g_N^\mu(K) \quad \text{for all } K.$$

By Lemma 8 each is concave with a unique zero, crossing from above, so a pointwise larger function has a weakly larger zero: $K_\nu^* \geq K_\mu^*$. Therefore $CRG(\nu) = m - NK_\nu^* \leq m - NK_\mu^* = CRG(\mu)$.

For the second claim, among distributions supported on $[0, \bar{v}]$ with mean E , the binary distribution $\underline{\mu}$ on $\{0, \bar{v}\}$ is the largest in the convex order: for any such μ and any convex ϕ , $\int \phi d\mu \leq \int \phi d\underline{\mu}$, because on $[0, \bar{v}]$ a convex function lies below the chord joining $(0, \phi(0))$ and $(\bar{v}, \phi(\bar{v}))$. Hence $CRG(\underline{\mu}) \leq CRG(\mu)$ for every such μ . $\square$

Proof of Corollary 1. For $\underline{\mu}$ we have $F^{-1}(t) = 0$ for $t < p_0 := 1 - E/\bar{v}$ and $F^{-1}(t) = \bar{v}$ for $t > p_0$, so $\Phi(q) = \bar{v}(q - p_0)^+$ and $q_K = p_0 + K/\bar{v}$. Then

$$\mathcal{R}(K) = \int_{q_K}^1 \frac{\bar{v}(q - p_0) - K}{q} dq = \bar{v} \int_{q_K}^1 \frac{q - q_K}{q} dq = \bar{v}[(1 - q_K) + q_K \log q_K].$$

Substituting $K = \bar{v}(q_K - p_0)$ and $E = \bar{v}(1 - p_0)$ into (10) and simplifying yields

$(N-1)(p_0 - q_K) = q_K \log q_K$, with $CRG = E - N\bar{v}(q_K - p_0)$. Uniqueness of the root is Lemma 8. As $N \rightarrow \infty$ we have $K^* \rightarrow 0$ and $q_K \rightarrow p_0$, so

$$CRG \rightarrow \bar{v}[(1 - p_0) + p_0 \log p_0] = E + (\bar{v} - E) \log \left(1 - \frac{E}{\bar{v}}\right). \quad \square$$

Remark 3. Because the binary prior is atomic, the density-form system (5)–(6) does not apply to it and the worst-case common bid must randomize. This is why Theorem 5 draws the bid through the uniform variable U rather than as a function of the value. In the limiting case the common bid at $v = \bar{v}$ has CDF $F_c(\delta) = \frac{\bar{v}-E}{E} \cdot \frac{\delta}{\bar{v}-\delta}$ on the support $[0, E]$.